claude-windows / 985cc286-9e1e-4217-87c3-2d418bc4fa9c

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\985cc286-9e1e-4217-87c3-2d418bc4fa9c\subagents\agent-aa1cce092daabbbf7.jsonl`
- SHA-256: `a07adc3e802e4909f436b9ddfae33e0d8a22f70fb2fd5129ac031472bfb621bc`
- Source modified: `2026-06-14T10:57:06+00:00`
- Imported at: `2026-07-05T16:48:25+00:00`
- project: `subagents`
- session_id: `985cc286-9e1e-4217-87c3-2d418bc4fa9c`

Transcript

1. user (2026-06-14T10:49:42.764Z)

URGENT (owner: "ensure we do not regress"): REVERT the Gate-A served-default flip (#146) ? it is a CONFIRMED served REGRESSION (eval?serve: ?0.008 served F1, drops ~12 real rallies; doubly-verified by an independent served check + PR #148). Restore the pre-#146 no-regression production state. Repo cwd C:/Users/avidu/Projects/badminton-highlight-indexer. ISOLATED worktree: `git -C ... fetch origin` then `git -C ... worktree add C:/Users/avidu/Projects/bhi-revert -b fix/revert-gate-a-default origin/master`.

Inspect `git show 570cf2b` (the #146 merge) for exactly what changed. REVERT the served-behavior changes: `config.json` `indexing.default_segmenter` back to its pre-#146 value (`gemini`); `backend/config/models.py` `FusionConfig.min_crossings` default back to `0` (Gate-A OFF). Fix the tests #146 added/changed (tests/test_fusion_hybrid.py, tests/test_config.py) back to the restored defaults (Gate-A OFF by default; the served default is the prior detector). Keep the persisted-superset invariant.

DOCS: amend docs/QUALITY_ITERATIONS.md so the Gate-A entry reads **REVERTED** with the honest eval?serve reason (the +0.074 was on the harness's recall-oriented PROPOSAL windowing; the served COARSE windowing makes Gate-A neutral/negative, ?0.008, dropping real rallies). A concurrent session may already have added a served-finding entry ? amend/consolidate, do NOT duplicate. Append a "reverted (eval?serve)" note to eval_baselines/experiment_leaderboard.json.

ALSO: the served-regression litmus PR #148 (branch `feat/served-gate-a-regression`) is the repeatable served check + runner that CAUGHT this ? green, additive verification infra. MERGE it if its CI is green and it's consistent with the revert (it documents the finding + guards against recurrence). If it conflicts, coordinate/rebase; goal = revert landed AND the served litmus preserved.

ruff+mypy+full pytest green. MERGE the revert on green (owner explicitly wants no regression ? this restores the known-good state). Branch-safe (explicit paths; `git log origin/master..HEAD` = only your commit). `Co-Authored-By: Claude Opus 4.8 (1M context) <noreply@anthropic.com>`. Return: revert PR number + merge status, #148 status, the restored config values, test results.

2. assistant (2026-06-14T10:49:45.020Z)

I'll start by reviewing the project conventions and then execute this revert carefully. Let me begin by pulling the latest and inspecting the #146 merge.

3. user (2026-06-14T10:49:48.016Z)

(Bash completed with no output)

4. user (2026-06-14T10:49:48.210Z)

commit 570cf2b1bc2455bc235a2b6eb0d60a09c28b833a
Author: Avi <avi.dullu@gmail.com>
Date: Sun Jun 14 15:51:42 2026 +0530

```
    feat(fusion): flip Gate-A (held-shuttle guard) to the served default ? config-reversible
(#146)

    Promote the validated Gate-A net-crossing guard (net_crossings >= 2) from
    "recommended config" to the live served default (owner-approved). Measured win:
    ?F1 +0.074, CI [+0.042,+0.105], 6/6 golden, sign-test p=0.031, over-seg 1.68->1.01;
    corroborated on the real GX010125 reel (45->27 rallies). Gate B stays OFF.

    Minimal, reversible change (rollback = flip a flag, no code change):
    - backend/config/models.py: FusionConfig.min_crossings default 0 -> 2 (Gate A on
      wherever fusion_hybrid serves). Rollback: indexing.fusion.min_crossings=0.
    - config.json: indexing.default_segmenter gemini -> fusion_hybrid (+ explicit
      indexing.fusion block making the served operating point visible). Rollback:
      indexing.default_segmenter=wasb_hybrid (full revert to the prior substrate).

    With default config FusionSegmenter reproduces the WASB substrate's seeding/handoff
    exactly except sub-2-crossing windows are dropped from the AI handoff. The persisted
    candidate superset is untouched (gating filters only the handoff), and the offline
    ablation harness is unaffected (it sets Variant.min_crossings explicitly, never
    reading FusionConfig).

    Tests (tests/test_config.py + tests/test_fusion_hybrid.py): default AppConfig applies
    Gate A; held-shuttle window (net_crossings<2) dropped by default while the real rally
    (>=2) is kept; reversibility (min_crossings=0 restores ungated handoff); persisted
    superset unchanged under the default gate; served default == fusion_hybrid. Full suite
    584 passed; ruff + mypy clean; coverage 80% (>=70 floor).

    Records the promotion: docs/QUALITY_ITERATIONS.md (PROMOTED-as-default + served-flip
    detail) and a promotion record in eval_baselines/experiment_leaderboard.json.

    Co-authored-by: Claude Opus 4.8 (1M context) <noreply@anthropic.com>

 backend/config/models.py                    | 33 +++++++++---
 config.json                                 |  8 ++-
 docs/QUALITY_ITERATIONS.md                  | 36 +++++++++++---
 eval_baselines/experiment_leaderboard.json | 45 ++++++++++++++++
 tests/test_config.py                        | 29 ++++++++--
 tests/test_fusion_hybrid.py                 | 86 ++++++++++++++++++++++++++++--
 6 files changed, 215 insertions(+), 22 deletions(-)
```

5. assistant (2026-06-14T10:49:50.731Z)

Let me look at the full diff of #146 and set up the isolated worktree.

6. user (2026-06-14T10:49:51.859Z)

```
commit 570cf2b1bc2455bc235a2b6eb0d60a09c28b833a
Author: Avi <avi.dullu@gmail.com>
Date:   Sun Jun 14 15:51:42 2026 +0530

    feat(fusion): flip Gate-A (held-shuttle guard) to the served default ? config-reversible
(#146)

    Promote the validated Gate-A net-crossing guard (net_crossings >= 2) from
    "recommended config" to the live served default (owner-approved). Measured win:
    ?F1 +0.074, CI [+0.042,+0.105], 6/6 golden, sign-test p=0.031, over-seg 1.68->1.01;
    corroborated on the real GX010125 reel (45->27 rallies). Gate B stays OFF.

    Minimal, reversible change (rollback = flip a flag, no code change):
    - backend/config/models.py: FusionConfig.min_crossings default 0 -> 2 (Gate A on
      wherever fusion_hybrid serves). Rollback: indexing.fusion.min_crossings=0.
    - config.json: indexing.default_segmenter gemini -> fusion_hybrid (+ explicit
      indexing.fusion block making the served operating point visible). Rollback:
      indexing.default_segmenter=wasb_hybrid (full revert to the prior substrate).
```

```
    With default config FusionSegmenter reproduces the WASB substrate's seeding/handoff
    exactly except sub-2-crossing windows are dropped from the AI handoff. The persisted
    candidate superset is untouched (gating filters only the handoff), and the offline
    ablation harness is unaffected (it sets Variant.min_crossings explicitly, never
    reading FusionConfig).

    Tests (tests/test_config.py + tests/test_fusion_hybrid.py): default AppConfig applies
    Gate A; held-shuttle window (net_crossings<2) dropped by default while the real rally
    (>=2) is kept; reversibility (min_crossings=0 restores ungated handoff); persisted
    superset unchanged under the default gate; served default == fusion_hybrid. Full suite
    584 passed; ruff + mypy clean; coverage 80% (>=70 floor).

    Records the promotion: docs/QUALITY_ITERATIONS.md (PROMOTED-as-default + served-flip
    detail) and a promotion record in eval_baselines/experiment_leaderboard.json.

    Co-authored-by: Claude Opus 4.8 (1M context) <noreply@anthropic.com>

diff --git a/backend/config/models.py b/backend/config/models.py
index ff94380..b18693e 100644
--- a/backend/config/models.py
+++ b/backend/config/models.py
@@ -78,12 +78,26 @@ class WasbIndexConfig(BaseModel):
 class FusionConfig(BaseModel):
     """Config for the multi-signal `fusion_hybrid` segmenter (MULTI_SIGNAL_FUSION_PLAN P2).

-    Every default reproduces control behaviour: the fusion segmenter persists the
-    shuttle net-crossing count (Gate A signal) and ? only when ``cue_flags['person']`` is
-    true ? the person-cue features, but it does NOT gate the served handoff unless a
-    threshold is raised (``min_crossings``/``min_players`` default 0 = OFF). The court mask
-    is optional: ``court_polygon=None`` ? Gate B counts every detection (player-count-only);
-    supplied ? off-court persons (spectators / adjacent courts) are excluded.
+    **Gate A (the held-shuttle guard) is ON by default** (``min_crossings=2``): a window
+    must show the shuttle crossing the net >= 2 times to reach the AI handoff, dropping the
+    held-shuttle / one-sided-feed / warm-up false positives. This is the promoted, validated
+    operating point (?F1 +0.074, 6/6 golden, sign-test p=0.031, CI [+0.042,+0.105]; over-seg
+    1.68?1.01 ? see docs/QUALITY_ITERATIONS.md). It only takes effect when the served segmenter
+    is ``fusion_hybrid`` (config.json ``indexing.default_segmenter``); the offline experiment
+    harness is unaffected (it sets ``Variant.min_crossings`` explicitly, never reading this).
+
+    **Reversible ? rollback is a single flag:** set ``indexing.fusion.min_crossings: 0`` to
+    restore the pre-Gate-A (ungated) handoff, or ``indexing.default_segmenter: wasb_hybrid``
+    to revert the served segmenter entirely. With ``min_crossings=0`` the fusion segmenter
+    reproduces the substrate (wasb_hybrid) byte-for-byte.
+
+    The person cue (Gate B) stays OFF by default: the fusion segmenter persists the shuttle
+    net-crossing count always, and ? only when ``cue_flags['person']`` is true ? the person-cue
+    features; ``min_players`` (Gate B) defaults to 0 = OFF. The court mask is optional:
+    ``court_polygon=None`` ? Gate B counts every detection (player-count-only); supplied ?
+    off-court persons (spectators / adjacent courts) are excluded. **The persisted candidate
+    set is the full superset regardless of any gate** ? gating only filters the AI handoff, so
+    the offline harness can re-score any variant.
     """
     # Which trajectory substrate seeds the windows (shuttle stays primary).
     substrate: Literal["wasb", "tracknet"] = "wasb"
@@ -95,8 +109,11 @@ class FusionConfig(BaseModel):
     # enabled ? so the default path never imports torch / touches the GPU.
     cue_flags: dict[str, bool] = Field(default_factory=dict)
     # Served Gate A: drop windows whose shuttle net-crossings < this from the AI handoff.
-    # 0 = OFF (no gating; persisted candidates are always the full superset for offline re-
scoring).
-    min_crossings: int = Field(default=0, ge=0)
+    # Default 2 = the PROMOTED held-shuttle guard (a real rally crosses the net >=2x; a single
+    # service feed / held shuttle crosses ~once). Set 0 to turn the gate OFF (ungated handoff;
+    # reproduces the substrate exactly). Persisted candidates are ALWAYS the full superset for
+    # offline re-scoring, regardless of this threshold.
```

```
+    min_crossings: int = Field(default=2, ge=0)
     # Served Gate B: when the person cue is on, drop windows with median in-court players
     # < this. 0 = OFF.
     min_players: int = Field(default=0, ge=0)
diff --git a/config.json b/config.json
index b72a93a..3107f70 100644
--- a/config.json
+++ b/config.json
@@ -23,7 +23,7 @@
         }
     },
     "indexing": {
-        "default_segmenter": "gemini",
+        "default_segmenter": "fusion_hybrid",
         "use_gpu": true,
         "motion_threshold": 0.08,
         "min_segment_duration": 3.0,
@@ -63,6 +63,12 @@
             "wsl_stage_dir": "~/clips",
             "timeout_sec": 1800,
             "velocity_threshold": 0.02
+        },
+        "fusion": {
+            "substrate": "wasb",
+            "min_crossings": 2,
+            "min_players": 0,
+            "cue_flags": {}
         }
     },
     "output": {
diff --git a/docs/QUALITY_ITERATIONS.md b/docs/QUALITY_ITERATIONS.md
index 308517d..12b6890 100644
--- a/docs/QUALITY_ITERATIONS.md
+++ b/docs/QUALITY_ITERATIONS.md
@@ -88,18 +88,42 @@ confirm it. **Next:** more matches (raise N until the sign test can resolve
~+0.
 *directly* (without the person handicap). Recorded in
`eval_baselines/experiment_leaderboard.json`. **Negative/uncertain
 results are kept on purpose ? this is the honest ablation backbone for the paper aim.**

+### Gate-A flipped to the SERVED DEFAULT (owner-approved) *(2026-06-14)*
+**Gate-A (net-crossings ? 2) is now the DEFAULT detection behaviour.** The owner approved
promoting the
+validated held-shuttle guard from "recommended config" to the live served default. The flip is
**two minimal,
+config-reversible changes**:
+- `backend/config/models.py` ? `FusionConfig.min_crossings` default **0 ? 2** (Gate A on
wherever
+  `fusion_hybrid` serves).
+- `config.json` ? `indexing.default_segmenter` **`gemini` ? `fusion_hybrid`** (+ an explicit
`indexing.fusion`
+  block making the served operating point visible). With default config, `FusionSegmenter`
reproduces the WASB
+  substrate's seeding/handoff exactly *except* that sub-2-crossing windows are dropped from the
AI handoff ? the
+  held-shuttle / one-sided-feed / warm-up false-positive fix.
+
+**Reversible ? rollback is a single flag** (no code change): set
`indexing.fusion.min_crossings: 0` to disable
+Gate A (ungated handoff, reproduces the substrate byte-for-byte), or
`indexing.default_segmenter: wasb_hybrid`
+to revert the served segmenter entirely. **Gate B stays OFF** (`min_players = 0`, person cue
OFF). **The
+persisted candidate superset is unchanged** ? gating filters only the AI handoff, so the
offline harness can
```

```
+still re-score any variant; the offline ablation harness is untouched (it sets
`Variant.min_crossings`
+explicitly and never reads `FusionConfig`). The validation evidence is unchanged from the
promotion below
+(?F1 +0.074, 6/6, p=0.031, CI [+0.042, +0.105], over-seg 1.68?1.01) plus the GX010125 45?27
corroboration.
+**Recorded:** a `promotion` record appended to `eval_baselines/experiment_leaderboard.json`. ?
**Honesty
+boundary:** the offline ?F1 is on the recall-oriented *proposal* candidate set, and the
production fusion
+segmenter's served path is still only mock-tested on real video (P2). The gate's effect is
**byte-reversible**,
+so this ships behind a one-flag rollback; a golden *served* regression on the fusion path
remains a worthwhile
+optional follow-up (owner-gated GPU+Gemini run) to confirm no served-side regression, but it
does not block the
+flip ? the offline 6/6 + p=0.031 ablation and the unit tests are the gate.
+
 ### Acting on the first A/B: Gate-A PROMOTED + learned variant FIXED + GX010125 corroboration
*(2026-06-14)*
 **Gate-A (net-crossings ? 2) ? PROMOTED (first cue promoted via the harness).** The measured
win
 (?F1 +0.074, **6/6 golden matches, sign-test p=0.031**, CI [+0.042,+0.104]; over-seg 1.68?1.01
? the
 [first-A/B archive](archives/quality-iterations/2026-06-first-ab-experiment/README.md)) clears
the honest
 bar. **Recommended config:** `indexing.fusion.min_crossings = 2` on the fusion path (the eval-
only Gate A
 folded into production, MULTI_SIGNAL_FUSION_PLAN §3.1). ? The offline win is on the recall-
oriented *proposal*
-candidate set; flipping the **live served default** stays owner-gated until a real fusion-path
run confirms it
-(the production fusion segmenter is P2/unverified on real video) ? so this records the
promotion + rationale,
-not a silent default flip. **Corroborated on real footage:** offline re-score of
GX010125_1080p's *cached*
-trajectory (no GPU, isolated `output/GX010125_run/`) ? 45 candidates ? CONTROL 45 rallies vs
**Gate-A 27**;
-Gate-A dropped **18 low-crossing windows (~1.7 min)** of likely held-shuttle/feed FPs (no
golden labels for
-this footage ? divergence, not F1).
+candidate set; flipping the **live served default** stayed owner-gated at the time of this
entry ? **now resolved:
+the owner approved the flip; see the SERVED-DEFAULT entry above.** **Corroborated on real
footage:** offline
+re-score of GX010125_1080p's *cached* trajectory (no GPU, isolated `output/GX010125_run/`) ? 45
candidates ?
+CONTROL 45 rallies vs **Gate-A 27**; Gate-A dropped **18 low-crossing windows (~1.7 min)** of
likely
+held-shuttle/feed FPs (no golden labels for this footage ? divergence, not F1).

 **Learned variant FIXED ? [#130](https://github.com/avidullu/badminton-highlight-
indexer/pull/130).** The
 first A/B's learned logistic lost to CONTROL and zeroed `testlarge` ? a fixed-0.5-cutoff
artifact. Added
diff --git a/eval_baselines/experiment_leaderboard.json
b/eval_baselines/experiment_leaderboard.json
index 610dcf8..b68bc77 100644
--- a/eval_baselines/experiment_leaderboard.json
+++ b/eval_baselines/experiment_leaderboard.json
@@ -90,5 +90,50 @@
         "p_value": 0.6875
       }
     }
+  },
```

```
+  {
+    "timestamp": "2026-06-14T00:00:00+00:00",
+    "iou": 0.5,
+    "label_set_hash": "50d98ffbc370",
+    "num_videos": 6,
+    "promotion": {
+      "cue": "gate_a_net_crossings",
+      "promoted_to": "served_default",
+      "decision": "PROMOTED ? Gate A (net_crossings>=2) flipped to the live served default",
+      "served_change": "config.json indexing.default_segmenter gemini->fusion_hybrid;
FusionConfig.min_crossings default 0->2",
+      "reversible": "indexing.fusion.min_crossings=0 (Gate-A rollback) or
indexing.default_segmenter=wasb_hybrid (full revert)",
+      "gate_b": "stays OFF (min_players=0, person cue OFF)",
+      "owner_approved": true,
+      "real_video_corroboration": "GX010125_1080p offline re-score: 45 candidates -> CONTROL 45
rallies vs Gate-A 27 (dropped 18 low-crossing held-shuttle/feed windows, ~1.7min)",
+      "over_seg_segment_count_ratio": {"before": 1.68, "after": 1.01},
+      "source_archive": "docs/archives/quality-iterations/2026-06-first-ab-
experiment/README.md"
+    },
+    "variant_a": {
+      "name": "control_shuttle_proposals",
+      "spec_hash": "control",
+      "aggregate": {}
+    },
+    "variant_b": {
+      "name": "gate_a_min_crossings_2",
+      "spec_hash": "gateA_mc2",
+      "aggregate": {}
+    },
+    "delta": {
+      "f1": 0.074
+    },
+    "honest_delta_f1": {
+      "mean_delta": 0.074,
+      "ci_low": 0.042,
+      "ci_high": 0.105,
+      "n": 6.0,
+      "ci_excludes_zero": true,
+      "sign_test": {
+        "wins": 6.0,
+        "losses": 0.0,
+        "ties": 0.0,
+        "n_effective": 6.0,
+        "p_value": 0.031
+      }
+    }
   }
 ]
\ No newline at end of file
diff --git a/tests/test_config.py b/tests/test_config.py
index fcb6033..6209841 100644
--- a/tests/test_config.py
+++ b/tests/test_config.py
@@ -40,18 +40,41 @@ def test_logging_config_file_override():
         assert cfg.logging.verbose_http is True


-def test_fusion_config_defaults_reproduce_control():
-    """The fusion block defaults must be no-ops: wasb substrate, no cues, no gating, no
mask."""
+def test_fusion_config_gate_a_on_by_default():
+    """Gate A (the held-shuttle guard) is ON by default at the promoted operating point
+    (min_crossings=2). Everything else stays a no-op: wasb substrate, no person cue, no
```

```
+    Gate B, no court mask."""
     cfg = load_config("/nonexistent/path/12345/config.json")
     f = cfg.indexing.fusion
     assert f.substrate == "wasb"
     assert f.cue_flags == {}            # person cue OFF
-    assert f.min_crossings == 0         # Gate A OFF
+    assert f.min_crossings == 2         # Gate A ON (PROMOTED default) ? the held-shuttle guard
     assert f.min_players == 0           # Gate B OFF
     assert f.court_polygon is None      # no mask
     assert f.velocity_threshold == 0.02


+def test_fusion_config_gate_a_reversible_to_zero():
+    """Rollback is a single flag: min_crossings=0 restores the pre-Gate-A ungated handoff
+    (the fusion segmenter then reproduces the wasb substrate byte-for-byte)."""
+    with tempfile.TemporaryDirectory() as td:
+        p = os.path.join(td, "rollback.json")
+        with open(p, "w") as fh:
+            json.dump({"indexing": {"fusion": {"min_crossings": 0}}}, fh)
+        cfg = load_config(p)
+        assert cfg.indexing.fusion.min_crossings == 0   # Gate A OFF ? explicit rollback
+
+
+def test_served_default_segmenter_is_fusion_hybrid():
+    """The shipped config.json flips the served default to fusion_hybrid so Gate A is the
+    live default detection behaviour (the project's bundled config, not the model fallback)."""
+    repo_root = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
+    cfg = load_config(os.path.join(repo_root, "config.json"))
+    assert cfg.indexing.default_segmenter == "fusion_hybrid"
+    # ...and the served fusion block has Gate A on at the promoted operating point.
+    assert cfg.indexing.fusion.min_crossings == 2
+
+
 def test_fusion_config_file_override():
     with tempfile.TemporaryDirectory() as td:
         p = os.path.join(td, "fusion.json")
diff --git a/tests/test_fusion_hybrid.py b/tests/test_fusion_hybrid.py
index 05d1f1d..27aae50 100644
--- a/tests/test_fusion_hybrid.py
+++ b/tests/test_fusion_hybrid.py
@@ -4,10 +4,16 @@ Proves: net_crossings is persisted; the person cue is OFF by default (no
person-
 keys, PersonDetector never built); the persisted candidate set is the full superset of the
 substrate windows; served Gate A drops sub-threshold windows from the AI handoff while
 persistence keeps the superset; and the person cue, when enabled, adds the person features.
+
+Also covers the PROMOTED Gate-A default (min_crossings=2): driven through a real AppConfig
+(the object main.py hands the segmenter), the held-shuttle window is dropped from the handoff
+by default, the real rally is kept, the persisted superset is unchanged, and min_crossings=0
+reverses it.
 """
 from types import SimpleNamespace
 from unittest.mock import MagicMock, patch

+from backend.config.models import AppConfig
 from backend.pipeline.detectors.fusion_features import PERSON_FEATURE_NAMES
 from backend.pipeline.segmenters.fusion_hybrid import FusionSegmenter

@@ -37,7 +43,16 @@ def _runner():
     return r


-def _run(indexing=None, provider_segments=None):
+def _run(indexing=None, provider_segments=None, config=None):
+    """Drive FusionSegmenter.process_video with a mocked runner/provider/db.
```

```
+
+    ``config`` (a real AppConfig or any object with ``.get``) takes precedence so a test can
+    exercise the SERVED path exactly as main.py builds it (config.get('indexing') -> the
+    Pydantic-defaulted dict, where fusion.min_crossings defaults to 2). When omitted, the
+    bare ``{"indexing": indexing or {}}`` dict is used ? that path has NO fusion block, so the
+    in-code ``.get("min_crossings", 0)`` fallback applies (ungated), which is intentional for
+    the older mechanics tests below.
+    """
     provider = MagicMock()
     provider.analyze_video.return_value = (provider_segments or [{"start_time": 0.5,
"end_time": 2.0}], {})
     db = MagicMock()
@@ -51,7 +66,7 @@ def _run(indexing=None, provider_segments=None):
          patch("backend.pipeline.segmenters.trajectory_hybrid.provider_registry") as preg, \
          patch("os.environ.get", return_value="dummy_key"):
         preg.get.return_value = provider
-        seg = FusionSegmenter(db, {"indexing": indexing or {}})
+        seg = FusionSegmenter(db, config if config is not None else {"indexing": indexing or
{}})
         res = seg.process_video("v1", "clip.mp4")
         return db, provider, res

@@ -82,8 +97,11 @@ def test_persisted_candidates_are_superset_of_substrate_windows():
     assert spans == [(1.0, 4.0), (10.0, 12.0)]          # exactly the substrate windows,
unfiltered


-def test_default_hands_off_all_windows():
-    _db, provider, _res = _run()                        # min_crossings default 0 -> no gating
+def test_no_fusion_block_hands_off_all_windows():
+    """A bare indexing dict with NO fusion block falls back to the in-code min_crossings=0
+    (ungated). This is the explicit-empty-config path, NOT the AppConfig default (which is
+    gated ? see the Gate-A-default tests below)."""
+    _db, provider, _res = _run()
     assert provider.analyze_video.call_count == 2


@@ -95,6 +113,66 @@ def test_gate_a_drops_low_crossing_window_from_handoff_only():
     assert provider.analyze_video.call_count == 1


+# --- Gate-A-as-default (the promotion under test) -----------------------------------------
+# These drive the SERVED path with a real AppConfig (the object main.py hands the segmenter),
+# so the Pydantic FusionConfig default (min_crossings=2) is in force.
+
+def test_appconfig_default_applies_gate_a_drops_held_shuttle():
+    """(a)+(b)+(c)+(e): with the DEFAULT AppConfig (no overrides), Gate A is on ? the
+    held-shuttle window (net_crossings<2) is DROPPED from the handoff while the real rally
+    (>=2) is KEPT, and the persisted candidate superset is unchanged (both windows
persisted)."""
+    db, provider, _res = _run(config=AppConfig())
+    # (e) persisted superset unchanged: both windows still written to candidate_segments.
+    spans = [(kw["start_time"], kw["end_time"]) for _a, kw in
db.add_candidate_segment.call_args_list]
+    assert spans == [(1.0, 4.0), (10.0, 12.0)]
+    # (b)+(c) only the high-crossing rally window A reaches the AI handoff; held window B
dropped.
+    assert provider.analyze_video.call_count == 1
+
+
+def test_appconfig_gate_a_keeps_real_rally():
+    """(c) sanity: a config whose ONLY window is the real rally still hands it off under the
+    default gate (the gate keeps >=2-crossing windows, it doesn't drop everything)."""
+    provider = MagicMock()
+    provider.analyze_video.return_value = ([{"start_time": 0.5, "end_time": 2.0}], {})
```

```
+    db = MagicMock()
+    db.add_candidate_segment.side_effect = [101, 102, 103, 104]
+    db.add_play_segment.return_value = 200
+    rally_only = MagicMock()
+    rally_only.healthcheck.return_value = (True, "env ok")
+    rally_only.run_predict.return_value = "out.csv"
+    rally_only.parse_trajectory_csv.return_value = _points()
+    rally_only.trajectory_to_action_windows.return_value = [dict(WIN_A)]   # rally only
+    with patch.object(FusionSegmenter, "_build_runner", return_value=(rally_only,
{"weights_path": "w"})), \
+         patch.object(FusionSegmenter, "_probe_fps_width", return_value=(30.0, 1280.0)), \
+         patch("backend.pipeline.segmenters.trajectory_hybrid.get_video_duration",
return_value=30.0), \
+         patch("backend.pipeline.segmenters.trajectory_hybrid.extract_video_chunk",
return_value=True), \
+         patch("backend.pipeline.segmenters.trajectory_hybrid.provider_registry") as preg, \
+         patch("os.environ.get", return_value="dummy_key"):
+        preg.get.return_value = provider
+        FusionSegmenter(db, AppConfig()).process_video("v1", "clip.mp4")
+    assert provider.analyze_video.call_count == 1       # the real rally is KEPT by default
+
+
+def test_gate_a_reversible_via_min_crossings_zero():
+    """(d) reversibility: min_crossings=0 restores the prior (ungated) behaviour even on the
+    AppConfig-driven served path ? BOTH windows reach the handoff again."""
+    cfg = AppConfig.model_validate({"indexing": {"fusion": {"min_crossings": 0}}})
+    assert cfg.indexing.fusion.min_crossings == 0
+    db, provider, _res = _run(config=cfg)
+    assert len(_stats_calls(db)) == 2                   # superset still persisted
+    assert provider.analyze_video.call_count == 2       # ungated: both windows handed off
+
+
+def test_gate_a_default_persists_full_superset():
+    """(e) explicit: the default gate filters ONLY the handoff ? every candidate window
+    (including the held-shuttle one it drops) is still persisted with its net_crossings
stat."""
+    db, _provider, _res = _run(config=AppConfig())
+    stats = _stats_calls(db)
+    assert len(stats) == 2                              # both persisted despite the gate
+    assert stats[0]["net_crossings"] >= 2               # rally
+    assert stats[1]["net_crossings"] == 0.0             # held shuttle ? persisted, just not
handed off
+
+
 def test_person_cue_adds_features_when_enabled():
     # Two in-court players, each shifting between two sampled frames ? exercises the REAL
     # glue (in-court counts -> players_in_court; track displacement -> mean_player_motion),
```

7. assistant (2026-06-14T10:49:54.839Z)

Now I have the full picture of #146. Let me check the current state of the relevant files on origin/master and look at PR #148, the experiment leaderboard, and check the pre-#146 value of `default_segmenter`.

8. user (2026-06-14T10:49:56.458Z)

```
e75ec35 feat(eval): LOSO/LOGO ablation runner ? per-signal marginal ?F1 (Phase 0) (#147)
570cf2b feat(fusion): flip Gate-A (held-shuttle guard) to the served default ? config-reversible
(#146)
7c6498f docs(quality): personalization Phase-0 checkpoint ? eval/safety rails + locked
structural-guard decision (#145)
4be16e1 docs(personalization): add PERSONALIZATION_PLAN ? hybrid per-user feature + locked owner
decisions (#144)
1a98ee8 feat(eval): per-user no-regression guard ? never make a user worse on a retrain/upgrade
(#143)
```

1b5af67 feat(eval): held-out-user learning-curve eval ? "<K videos to superior quality" as a release gate (#142)
7376415 feat(eval): per-user promotion gate for flag-gated personalization (#141)
77082f4 feat(eval): wire run-telemetry black box into fusion_golden extractor (#140)
ec5bbe0 chore(eval): track experiment_leaderboard.json (fusion P3/P4 honest LOVO records) (#139)
74c1303 docs(quality): log fusion P3+P4 measured ?F1 on the n=6 golden corpus (#137)
8836dfe chore: scripts/ for run-drivers + CODE_MAP §0 file-placement guidelines (#138)
fa12829 chore: gitignore scratch/ to keep git status clean (#136)
55df82f feat(obs): run-telemetry black box ? fps + GPU temp/clocks/power + CPU/RAM (rotating ~10KB JSONL) (#134)
a9432a9 feat(db): self-scheduling job substrate ? migration v3 + claim_due_job (P2) (#135)
ec2002d feat(infra): ExecutionPolicy P1 ? bound the Gemini hang (op_timeout + bounded poll) (#133)

9. user (2026-06-14T10:49:57.407Z)

26:        "default_segmenter": "gemini",

10. user (2026-06-14T10:50:00.370Z)

{"baseRefName":"master","headRefName":"feat/served-gate-a-regression","mergeStateStatus":"CLEAN","mergeable":"MERGEABLE","number":148,"state":"OPEN","title":"feat(eval): served Gate-A verification + regression litmus ? DIVERGENCE finding (#146 follow-up)"}
---PR146---
{"mergedAt":"2026-06-14T10:21:42Z","number":146,"state":"MERGED","title":"feat(fusion): flip Gate-A (held-shuttle guard) to the served default ? config-reversible"}

11. assistant (2026-06-14T10:50:03.349Z)

Pre-#146 `default_segmenter` value is confirmed as `gemini`. PR #148 is OPEN, CLEAN/MERGEABLE, based on master. Let me look at PR #148's contents and check if a concurrent session already added a served-finding entry to the docs and leaderboard.

12. user (2026-06-14T10:50:06.096Z)

{"body":"## What\n\nCloses the flagged gap from #146: the served Gate-A default flip\n(`default_segmenter=fusion_hybrid` + `min_crossings=2`) was **mock-tested only** on real\nvideo. This runs the **real** `FusionSegmenter` decision seams (`_enrich_candidate_stats`\nnet-crossings + `_select_for_handoff` Gate A) over the **6 cached golden WASB trajectories**\nat the **production windowing operating point** ? GPU-free, Gemini-free (person cue OFF; it\nstops at the AI-handoff boundary).\n\n- `backend/eval/served_gate_a.py` ? runner helper. Production `FusionSegmenter` handoff\n selection on a cached trajectory CSV, scored vs golden labels (F1 + over-seg ratio),\n Gate-A ON (`min_crossings=2`) vs OFF (`0`). CLI + library API.\n- `tests/test_served_gate_a_regression.py` ? repeatable litmus. **Skips cleanly** when the\n golden trajectories are absent (CI), **runs on the owner box** (mirrors the ablation\n litmus). `SERVED_GATE_A_MANIFEST` env var points it at an out-of-tree checkout's manifest.\n\n## ? Verdict: DIVERGENCE ? Gate A does NOT reproduce the +0.074 on the served path\n\nOn the **production windowing** Gate A is **~neutral, not +0.074**:\n\n| windowing | mean ?F1 | wins | over-seg (off?on) |\n|---|---|---|---|\n| **served / production** (`vt=0.02, gap=2.0, min=2.0`) | **?0.008** | 2/6 (2 losses) | 0.70 ? 0.62 |\n| offline **PROPOSAL** (`vt=0.005, gap=1.0, min=0.5`) | **+0.074** | 6/6 (p=0.031) | 1.68 ? 1.01 |\n\nServed per-video (production windowing, IoU?0.5):\n\n```\nvideo   gt F1_off F1_on   ?F1 segR_off segR_on\nadarsh_avi_singles   96   0.632 0.628 -0.004   1.01   0.79\nkushagra_singles 32   0.000 0.000 +0.000   0.06   0.06\nlargetest_doubles   49   0.535 0.484 -0.051 1.06   0.86\ngbaaddy   48   0.086 0.088 +0.003   0.46   0.42\ntestlarge_short 6   0.182 0.182 +0.000   0.83   0.83\nmahadevpura_singles   31   0.364 0.370 +0.007 0.77   0.74\nMEAN   0.300 0.292 -0.008\n```\n\n### Root cause (isolated, not a bug)\n\nThe +0.074 is a property of the **offline harness's permissive proposal windowing**, not of\nGate A in isolation. The control test `test_proposal_windowing_reproduces_offline_lift`\nreproduces +0.074 / 6-of-6 / over-seg 1.68?1.01 **exactly** by applying the *same*\n`rally_gate` net-crossing gate to proposal-windowed candidates ? so the gate is correct; the\noperating point is what differs.\n\nProduction windowing (`vt=0.02, gap=2.0, min=2.0`) is already conservative: it merges flight\nbursts into few long windows and drops short ones, producing **far fewer candidates** than\nthe permissive proposal ? e.g. gbaaddy **22 vs 82**, kushagra **2 vs 20**. The served over-seg\nratio is already **< 1** (under-segmentation), so there is little held-shuttle junk left for\nGate A to cut; it occasionally trims a borderline true window instead ? slightly negative ?F1.\n\n## Why this is open for review, not merged\n\nPer the verification brief: do not merge a misleading \"pass\" on a real divergence. The flip is\n**reversible** (`min_crossings: 0` or `default_segmenter: wasb_hybrid`). Owner decision needed ?\noptions:\n1. **Keep the flip** ? Gate A is ~neutral on the served path (no real regression: mean ?F1\n ?0.008, within noise; never collapses F1), and it's a structurally-motivated guard that\n   pays off on footage with more held-shuttle service feeds than the golden 6.\n2. **Align served windowing toward the proposal operating point** so the measured +0.074\n   transfers ? then re-run this litmus to confirm.\n3. **Revert the served default** until (2) lands.\n\nThe test asserts the **honest measured** behaviour (served ~neutral; +0.074 only under\nproposal windowing), so whichever path is chosen, this litmus pins the truth and trips if a\nfuture windowing change silently revives or breaks either fact.\n\n## Verify\n- `ruff check` + `mypy backend/eval/` clean.\n- Full suite: **609 passed, 7 skipped** (golden-data-gated skips).\n- New litmus: 5 passed on the owner box (golden data present), 5 skipped without it.\n\n? Generated with [Claude Code](https://claude.com/claude-code)\n","commits":[{"authoredDate":"2026-06-14T10:44:00Z","authors":[{"email":"avi.dullu@gmail.com","id":"MDQ6VXNlcjE2NzQ4Nzc=","login":"avidullu","name":"Avi Dullu"},{"email":"noreply@anthropic.com","id":"MDQ6VXNlcjgxODQ3","login":"claude","name":"Claude Opus 4.8 (1M context)"}],"committedDate":"2026-06-14T10:44:00Z","messageBody":"? (#146 follow-up)\n\nCloses the flagged gap: the served default flip (default_segmenter=fusion_hybrid +\nmin_crossings=2, #146) was mock-tested only on real video. This drives the REAL\nFusionSegmenter decision seams (_enrich_candidate_stats net-crossings +\n_select_for_handoff Gate A) over the 6 cached golden WASB trajectories at the\nPRODUCTION windowing operating point ? GPU-free, Gemini-free (person cue OFF).\n\nbackend/eval/served_gate_a.py ? runner helper: production FusionSegmenter handoff\nselection on a cached trajectory CSV, scored vs golden labels (F1 + over-seg ratio),\nGate-A ON (min_crossings=2) vs OFF (0). CLI + library API.\n\ntests/test_served_gate_a_regression.py ? skips cleanly when golden trajectories are\nabsent (CI), runs on the owner box (like the ablation litmus). Asserts the served\nno-regression + monotone-over-seg invariants, AND a control test proving the +0.074\nlift reproduces under the offline PROPOSAL windowing.\n\nHONEST FINDING (verdict: DIVERGENCE, not a clean HOLD): on the PRODUCTION windowing\nGate A is ~neutral (mean served ?F1 ? -0.008, over-seg already 0.62-0.70 < 1), NOT\nthe +0.074 the offline ablation reports. Root cause isolated to windowing: production\nconfig (velocity_threshold=0.02, merge_gap=2.0, min_window=2.0) yields far fewer, longer\ncandidate windows than the permissive offline PROPOSAL (0.005, 1.0, 0.5) ? e.g. gbaaddy\n22 vs 82 windows ? so there is little held-shuttle junk left for Gate A to cut. The\n+0.074 reproduces exactly (6/6, p=0.031, over-seg 1.68->1.01) ONLY under PROPOSAL\nwindowing. The flip is reversible (min_crossings=0 / default_segmenter=wasb_hybrid).\n\nCo-Authored-By: Claude Opus 4.8 (1M context) <noreply@anthropic.com>","messageHeadline":"feat(eval): served Gate-A verification + repeatable regression litmus?","oid":"fd0d7b3ea214b5c141ea53a986ffe780a22396e4"}],"files":[{"path":"backend/eval/served_gate_a.py","additions":199,"deletions":0,"changeType":"ADDED"},{"path":"tests/test_served_gate_a_regression.py","additions":160,"deletions":0,"changeType":"ADDED"}]}

13. user (2026-06-14T10:50:07.202Z)

Full test suite (offline, mocked)   pass   2m2s   https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27496356623/job/81271029709
Mypy (lenient baseline) pass   32s   https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27496356623/job/81271029652
Ruff lint   pass   9s   https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27496356623/job/81271029679
Syntax sweep + import smoke (no GPU stack)   pass   17s   https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27496356623/job/81271029666

14. assistant (2026-06-14T10:50:10.810Z)

PR #148 is green, CLEAN/MERGEABLE, and additive (only adds `backend/eval/served_gate_a.py` + `tests/test_served_gate_a_regression.py` ? no conflict with my revert files). The verdict (?0.008, divergence) corroborates the revert.

Now let me set up the isolated worktree and inspect the current state of the files I need to revert.

15. user (2026-06-14T10:50:12.291Z)

```
Preparing worktree (new branch 'fix/revert-gate-a-default')
branch 'fix/revert-gate-a-default' set up to track 'origin/master'.
HEAD is now at e75ec35 feat(eval): LOSO/LOGO ablation runner ? per-signal marginal ?F1 (Phase 0) (#147)
```

16. user (2026-06-14T10:50:16.229Z)

```
fix/revert-gate-a-default
---HEAD---
e75ec35 feat(eval): LOSO/LOGO ablation runner ? per-signal marginal ?F1 (Phase 0) (#147)
---only my commits check (should be empty now)---
```

17. assistant (2026-06-14T10:50:19.670Z)

Worktree is clean and based on origin/master. Now let me read the current state of the files I need to modify.

18. user (2026-06-14T10:50:20.645Z)

```
        # Warn before a run if WSL free space (at wsl_stage_dir) is below this many GB. 0 = disabled.
        min_free_gb: float = 0.0


    class FusionConfig(BaseModel):
        """Config for the multi-signal `fusion_hybrid` segmenter (MULTI_SIGNAL_FUSION_PLAN P2).

        **Gate A (the held-shuttle guard) is ON by default** (``min_crossings=2``): a window
        must show the shuttle crossing the net >= 2 times to reach the AI handoff, dropping the
        held-shuttle / one-sided-feed / warm-up false positives. This is the promoted, validated
        operating point (?F1 +0.074, 6/6 golden, sign-test p=0.031, CI [+0.042,+0.105]; over-seg
        1.68?1.01 ? see docs/QUALITY_ITERATIONS.md). It only takes effect when the served segmenter
        is ``fusion_hybrid`` (config.json ``indexing.default_segmenter``); the offline experiment
        harness is unaffected (it sets ``Variant.min_crossings`` explicitly, never reading this).

        **Reversible ? rollback is a single flag:** set ``indexing.fusion.min_crossings: 0`` to
        restore the pre-Gate-A (ungated) handoff, or ``indexing.default_segmenter: wasb_hybrid``
        to revert the served segmenter entirely. With ``min_crossings=0`` the fusion segmenter
        reproduces the substrate (wasb_hybrid) byte-for-byte.

        The person cue (Gate B) stays OFF by default: the fusion segmenter persists the shuttle
        net-crossing count always, and ? only when ``cue_flags['person']`` is true ? the person-cue
        features; ``min_players`` (Gate B) defaults to 0 = OFF. The court mask is optional:
        ``court_polygon=None`` ? Gate B counts every detection (player-count-only); supplied ?
        off-court persons (spectators / adjacent courts) are excluded. **The persisted candidate
        set is the full superset regardless of any gate** ? gating only filters the AI
```

```
    handoff, so
    the offline harness can re-score any variant.
    """
    # Which trajectory substrate seeds the windows (shuttle stays primary).
    substrate: Literal["wasb", "tracknet"] = "wasb"
    # Windowing velocity threshold for the fusion family (the engine reads
    # indexing.<family>.velocity_threshold). Default matches the substrates' 0.02; set
    # equal to indexing.wasb.velocity_threshold to A/B fusion against a tuned wasb run.
    velocity_threshold: float = 0.02
    # Per-cue enable flags, e.g. {"person": true}. Person cue is OFF unless explicitly
    # enabled ? so the default path never imports torch / touches the GPU.
    cue_flags: dict[str, bool] = Field(default_factory=dict)
    # Served Gate A: drop windows whose shuttle net-crossings < this from the AI handoff.
    # Default 2 = the PROMOTED held-shuttle guard (a real rally crosses the net >=2x; a single
    # service feed / held shuttle crosses ~once). Set 0 to turn the gate OFF (ungated handoff;
    # reproduces the substrate exactly). Persisted candidates are ALWAYS the full superset for
    # offline re-scoring, regardless of this threshold.
    min_crossings: int = Field(default=2, ge=0)
    # Served Gate B: when the person cue is on, drop windows with median in-court players
    # < this. 0 = OFF.
    min_players: int = Field(default=0, ge=0)
    # Optional court mask as normalised 0-1 [[x, y], ...] polygon. None = no mask.
    court_polygon: Optional[list[list[float]]] = None
    # Net line for the person two-sided-motion split (person features only ? the shuttle
    # net-crossing gate keeps rally_gate's camera-agnostic auto-estimate).
    net_axis: Literal["x", "y"] = "y"
    net_position: float = Field(default=0.5, ge=0.0, le=1.0)


class IndexingConfig(BaseModel):
    default_segmenter: str = "yolo_hybrid"
    use_gpu: bool = True
    motion_threshold: float = 0.08
    min_segment_duration: float = 3.0
    dead_time_merge_gap: float = 2.0
```

19. user (2026-06-14T10:50:21.021Z)


```
{
    "sport": "badminton",
    "ai_provider": "gemini",
    "ai_model": "gemini-flash-latest",
    "validation": {
        "min_resolution_width": 1280,
        "min_resolution_height": 720,
        "min_fps": 29.9,
        "min_blur_threshold": 20.0,
        "max_scene_cuts_per_minute": 0.5,
        "relevance": {
            "court_green_lower": [
                35,
                40,
                40
            ],
            "court_green_upper": [
                85,
                255,
                255
            ],
            "court_pixels_min_ratio": 0.05
```

```
            }
        },
        "indexing": {
            "default_segmenter": "fusion_hybrid",
            "use_gpu": true,
            "motion_threshold": 0.08,
            "min_segment_duration": 3.0,
            "dead_time_merge_gap": 2.0,
            "chunk_duration": 90,
            "chunk_overlap": 10,
            "detector_model": "fasterrcnn_resnet50_fpn",
            "detector_score_threshold": 0.5,
            "yolo_velocity_threshold": 0.15,
            "yolo_frame_skip": 3,
            "yolo_tracker": "bytetrack.yaml",
            "yolo_prefetch_workers": 2,
            "min_action_window_duration": 2.0,
            "log_yolo_candidates": true,
            "store_yolo_candidates": true,
            "ai_handoff_padding": 2.0,
            "ai_max_workers": 1,
            "tracknet": {
                "wsl_distro": "Ubuntu",
                "repo_dir": "~/models/TrackNetV4",
                "conda_sh": "~/miniconda3/etc/profile.d/conda.sh",
                "conda_env": "TrackNetV4",
                "weights_path": "",
                "queue_length": 5,
                "stage_in_wsl": true,
                "wsl_stage_dir": "~/clips",
                "timeout_sec": 1800,
                "velocity_threshold": 0.02
            },
            "wasb": {
                "wsl_distro": "Ubuntu",
                "repo_dir": "~/models/WASB-SBDT",
                "conda_sh": "~/miniconda3/etc/profile.d/conda.sh",
                "conda_env": "wasb",
                "weights_path": "~/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar",
                "sport": "badminton",
                "wsl_stage_dir": "~/clips",
                "timeout_sec": 1800,
                "velocity_threshold": 0.02
            },
            "fusion": {
                "substrate": "wasb",
                "min_crossings": 2,
                "min_players": 0,
                "cue_flags": {}
            }
        },
        "output": {
            "default_highlights_name": "highlights.mp4",
            "db_name": "sports_indexer.db"
        },
        "stitching": {
            "begin_padding": 0.5,
            "end_padding": 0.5,
            "use_cache": false,
            "min_shot_count": 1,
            "watermark": {
                "enabled": true,
                "text": "Generated by Khelsutra.guru",
                "position": "bottom-right",
```

```
                "opacity": 0.85,
                "logo_path": "",
                "font_path": ""
            }
        }
    }
```

20. assistant (2026-06-14T10:50:30.172Z)

Now I'll make the reverts. First, `config.json`: restore `default_segmenter` to `gemini`. Note that #146 added an `indexing.fusion` block that did not exist pre-#146. Let me verify whether that block existed before #146.

21. user (2026-06-14T10:50:31.947Z)

```
        },
        "wasb": {
            "wsl_distro": "Ubuntu",
            "repo_dir": "~/models/WASB-SBDT",
            "conda_sh": "~/miniconda3/etc/profile.d/conda.sh",
            "conda_env": "wasb",
            "weights_path": "~/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar",
            "sport": "badminton",
            "wsl_stage_dir": "~/clips",
            "timeout_sec": 1800,
            "velocity_threshold": 0.02
        }
    },
    "output": {
        "default_highlights_name": "highlights.mp4",
        "db_name": "sports_indexer.db"
    },
    "stitching": {
        "begin_padding": 0.5,
        "end_padding": 0.5,
        "use_cache": false,
```

22. assistant (2026-06-14T10:50:37.844Z)

Confirmed: pre-#146 there was no `indexing.fusion` block in config.json. The brief specifies restoring `default_segmenter` to `gemini` and `FusionConfig.min_crossings` default back to `0`.

For the `indexing.fusion` block in config.json: with the model default back at `min_crossings=0`, this block now explicitly sets `min_crossings: 2` ? which would keep Gate A ON for fusion_hybrid even though it's no longer the served default. Since the goal is to restore the pre-#146 no-regression state and Gate-A is confirmed neutral/negative on serving, I'll remove this added block entirely to match the pre-#146 state exactly (full revert of the served-behavior changes). Let me make the edits.

23. user (2026-06-14T10:50:39.313Z)

The file C:/Users/avidu/Projects/bhi-revert/config.json has been updated successfully. (file state is current in your context ? no need to Read it back)

24. user (2026-06-14T10:50:41.079Z)

The file C:/Users/avidu/Projects/bhi-revert/config.json has been updated successfully. (file state is current in your context ? no need to Read it back)

25. assistant (2026-06-14T10:50:44.308Z)

Now revert `backend/config/models.py`: restore the `FusionConfig` docstring and `min_crossings` default to `0`. Let me restore the pre-#146 docstring exactly and flip the default.

26. user (2026-06-14T10:50:54.385Z)

```
The file C:/Users/avidu/Projects/bhi-revert/backend/config/models.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
```

27. user (2026-06-14T10:50:59.874Z)

```
The file C:/Users/avidu/Projects/bhi-revert/backend/config/models.py has been updated
successfully. (file state is current in your context ? no need to Read it back)
```

28. assistant (2026-06-14T10:51:02.184Z)

```
Now let me verify the models.py changes match the pre-#146 state exactly with a diff against
`570cf2b^`.
```

29. user (2026-06-14T10:51:04.458Z)

```
(Bash completed with no output)
```

30. assistant (2026-06-14T10:51:07.961Z)

```
`backend/config/models.py` and `config.json` now match the pre-#146 state byte-for-byte. Now let
me fix the tests. First the test files.
```

31. user (2026-06-14T10:51:08.559Z)

```
        assert cfg.logging.verbose_http is True


def test_fusion_config_gate_a_on_by_default():
    """Gate A (the held-shuttle guard) is ON by default at the promoted operating point
    (min_crossings=2). Everything else stays a no-op: wasb substrate, no person cue, no
    Gate B, no court mask."""
    cfg = load_config("/nonexistent/path/12345/config.json")
    f = cfg.indexing.fusion
    assert f.substrate == "wasb"
    assert f.cue_flags == {}            # person cue OFF
    assert f.min_crossings == 2         # Gate A ON (PROMOTED default) ? the held-shuttle guard
    assert f.min_players == 0           # Gate B OFF
    assert f.court_polygon is None      # no mask
    assert f.velocity_threshold == 0.02


def test_fusion_config_gate_a_reversible_to_zero():
    """Rollback is a single flag: min_crossings=0 restores the pre-Gate-A ungated handoff
    (the fusion segmenter then reproduces the wasb substrate byte-for-byte)."""
    with tempfile.TemporaryDirectory() as td:
        p = os.path.join(td, "rollback.json")
        with open(p, "w") as fh:
            json.dump({"indexing": {"fusion": {"min_crossings": 0}}}, fh)
        cfg = load_config(p)
        assert cfg.indexing.fusion.min_crossings == 0   # Gate A OFF ? explicit rollback


def test_served_default_segmenter_is_fusion_hybrid():
    """The shipped config.json flips the served default to fusion_hybrid so Gate A is the
    live default detection behaviour (the project's bundled config, not the model fallback)."""
    repo_root = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
    cfg = load_config(os.path.join(repo_root, "config.json"))
    assert cfg.indexing.default_segmenter == "fusion_hybrid"
```

```
    # ...and the served fusion block has Gate A on at the promoted operating point.
    assert cfg.indexing.fusion.min_crossings == 2


def test_fusion_config_file_override():
    with tempfile.TemporaryDirectory() as td:
        p = os.path.join(td, "fusion.json")
        with open(p, "w") as fh:
            json.dump({"indexing": {"fusion": {
                "substrate": "tracknet", "min_crossings": 2,
                "cue_flags": {"person": True}, "court_polygon": [[0.1, 0.1], [0.9, 0.1],
[0.5, 0.9]],
```

32. assistant (2026-06-14T10:51:12.000Z)

Let me look at what the pre-#146 version of these tests looked like, to restore them faithfully.

33. user (2026-06-14T10:51:13.805Z)

```
        assert cfg.logging.verbose_http is True


def test_fusion_config_defaults_reproduce_control():
    """The fusion block defaults must be no-ops: wasb substrate, no cues, no gating, no mask."""
    cfg = load_config("/nonexistent/path/12345/config.json")
    f = cfg.indexing.fusion
    assert f.substrate == "wasb"
    assert f.cue_flags == {}            # person cue OFF
    assert f.min_crossings == 0         # Gate A OFF
    assert f.min_players == 0           # Gate B OFF
    assert f.court_polygon is None      # no mask
    assert f.velocity_threshold == 0.02


def test_fusion_config_file_override():
    with tempfile.TemporaryDirectory() as td:
        p = os.path.join(td, "fusion.json")
        with open(p, "w") as fh:
            json.dump({"indexing": {"fusion": {
                "substrate": "tracknet", "min_crossings": 2,
                "cue_flags": {"person": True}, "court_polygon": [[0.1, 0.1], [0.9, 0.1], [0.5,
0.9]],
            }}}, fh)
        cfg = load_config(p)
        f = cfg.indexing.fusion
        assert f.substrate == "tracknet"
        assert f.min_crossings == 2
        assert f.cue_flags == {"person": True}
        assert len(f.court_polygon) == 3
        # An old config without a fusion block still loads (defaults applied).
        assert cfg.indexing.wasb.velocity_threshold == 0.02


def test_file_override():
    with tempfile.TemporaryDirectory() as td:
        p = os.path.join(td, "override.json")
        data = {
            "indexing": {
                "default_segmenter": "gemini",
                "ai_max_workers": 3,
            },
```

34. assistant (2026-06-14T10:51:20.850Z)

The pre-#146 version had only `test_fusion_config_defaults_reproduce_control` (no Gate-A-on test, no reversibility test, no served-default test). The brief says "Fix the tests #146 added/changed back to the restored defaults (Gate-A OFF by default; the served default is the prior detector)."

I'll restore `test_fusion_config_defaults_reproduce_control` to its pre-#146 form, and update the two added tests rather than delete them outright ? they provide useful coverage of the restored state (reversibility/explicit-zero is still valid; the served-default test should now assert `gemini`). Let me replace the #146 block in test_config.py.

35. user (2026-06-14T10:51:35.470Z)

The file C:/Users/avidu/Projects/bhi-revert/tests/test_config.py has been updated successfully. (file state is current in your context ? no need to Read it back)

36. assistant (2026-06-14T10:51:40.490Z)

Now the test_fusion_hybrid.py. Let me read the current full file to understand the structure of the Gate-A-default tests added by #146.

37. user (2026-06-14T10:51:40.865Z)

```
"""FusionSegmenter tests (mocked runner/provider/db ? no GPU/WSL).

Proves: net_crossings is persisted; the person cue is OFF by default (no person-feature
keys, PersonDetector never built); the persisted candidate set is the full superset of the
substrate windows; served Gate A drops sub-threshold windows from the AI handoff while
persistence keeps the superset; and the person cue, when enabled, adds the person features.

Also covers the PROMOTED Gate-A default (min_crossings=2): driven through a real AppConfig
(the object main.py hands the segmenter), the held-shuttle window is dropped from the handoff
by default, the real rally is kept, the persisted superset is unchanged, and min_crossings=0
reverses it.
"""
from types import SimpleNamespace
from unittest.mock import MagicMock, patch

from backend.config.models import AppConfig
from backend.pipeline.detectors.fusion_features import PERSON_FEATURE_NAMES
from backend.pipeline.segmenters.fusion_hybrid import FusionSegmenter

# Window A [1,4]s: shuttle alternates across the net every frame -> many crossings (a rally).
# Window B [10,12]s: shuttle sits at the net level (held) -> 0 crossings. fps=30.
WIN_A = {"start": 1.0, "end": 4.0, "active_frame_count": 90, "peak_velocity": 0.4,
         "mean_velocity": 0.2, "track_id_count": 2, "chunk_index": 0}
WIN_B = {"start": 10.0, "end": 12.0, "active_frame_count": 60, "peak_velocity": 0.1,
         "mean_velocity": 0.05, "track_id_count": 1, "chunk_index": 0}


def _points():
    pts = []
    for f in range(30, 120):       # window A frames: y flips 100<->900 around the net (~500)
        pts.append(SimpleNamespace(frame=f, visible=True, x=500.0, y=100.0 if f % 2 == 0 else 900.0))
    for f in range(300, 360):      # window B frames: y == net level -> deadband -> no crossings
        pts.append(SimpleNamespace(frame=f, visible=True, x=500.0, y=500.0))
```

```
    return pts


def _runner():
    r = MagicMock()
    r.healthcheck.return_value = (True, "env ok")
    r.run_predict.return_value = "out.csv"
    r.parse_trajectory_csv.return_value = _points()
    r.trajectory_to_action_windows.return_value = [dict(WIN_A), dict(WIN_B)]
    return r


def _run(indexing=None, provider_segments=None, config=None):
    """Drive FusionSegmenter.process_video with a mocked runner/provider/db.

    ``config`` (a real AppConfig or any object with ``.get``) takes precedence so a test can
    exercise the SERVED path exactly as main.py builds it (config.get('indexing') -> the
    Pydantic-defaulted dict, where fusion.min_crossings defaults to 2). When omitted, the
    bare ``{"indexing": indexing or {}}`` dict is used ? that path has NO fusion block, so the
    in-code ``.get("min_crossings", 0)`` fallback applies (ungated), which is intentional for
    the older mechanics tests below.
    """
    provider = MagicMock()
    provider.analyze_video.return_value = (provider_segments or [{"start_time": 0.5, "end_time": 2.0}], {})
    db = MagicMock()
    db.add_candidate_segment.side_effect = [101, 102, 103, 104]
    db.add_play_segment.return_value = 200

    with patch.object(FusionSegmenter, "_build_runner", return_value=(_runner(), {"weights_path": "w"})), \
         patch.object(FusionSegmenter, "_probe_fps_width", return_value=(30.0, 1280.0)), \
         patch("backend.pipeline.segmenters.trajectory_hybrid.get_video_duration", return_value=30.0), \
         patch("backend.pipeline.segmenters.trajectory_hybrid.extract_video_chunk", return_value=True), \
         patch("backend.pipeline.segmenters.trajectory_hybrid.provider_registry") as preg, \
         patch("os.environ.get", return_value="dummy_key"):
        preg.get.return_value = provider
        seg = FusionSegmenter(db, config if config is not None else {"indexing": indexing or {}})
        res = seg.process_video("v1", "clip.mp4")
        return db, provider, res


def _stats_calls(db):
    """Persisted stats dicts, in call order."""
    return [kw["stats"] for _a, kw in db.add_candidate_segment.call_args_list]


def test_net_crossings_persisted_and_person_off_by_default():
    db, _provider, _res = _run()
    stats = _stats_calls(db)
    assert len(stats) == 2                              # both windows persisted (superset)
    assert all("net_crossings" in s for s in stats)
    # Window A (rally) crosses many times; window B (held) zero.
    assert stats[0]["net_crossings"] >= 2
    assert stats[1]["net_crossings"] == 0.0
```

```
        # Person cue OFF by default -> NONE of the person features present.
        assert all(not any(k in s for k in PERSON_FEATURE_NAMES) for s in stats)
        # The 4 base motion keys are still there.
        assert all({"peak_velocity", "mean_velocity", "active_frame_count", "track_id_count"} <= set(s)
                   for s in stats)


    def test_persisted_candidates_are_superset_of_substrate_windows():
        db, _provider, _res = _run()
        spans = [(kw["start_time"], kw["end_time"]) for _a, kw in db.add_candidate_segment.call_args_list]
        assert spans == [(1.0, 4.0), (10.0, 12.0)]          # exactly the substrate windows, unfiltered


    def test_no_fusion_block_hands_off_all_windows():
        """A bare indexing dict with NO fusion block falls back to the in-code min_crossings=0
        (ungated). This is the explicit-empty-config path, NOT the AppConfig default (which is
        gated ? see the Gate-A-default tests below)."""
        _db, provider, _res = _run()
        assert provider.analyze_video.call_count == 2


    def test_gate_a_drops_low_crossing_window_from_handoff_only():
        db, provider, _res = _run(indexing={"fusion": {"min_crossings": 2}})
        # Persistence is still the full superset (offline substrate intact)...
        assert len(_stats_calls(db)) == 2
        # ...but only the high-crossing window A reaches the AI handoff.
        assert provider.analyze_video.call_count == 1


    # --- Gate-A-as-default (the promotion under test) -----------------------------------------
    # These drive the SERVED path with a real AppConfig (the object main.py hands the segmenter),
    # so the Pydantic FusionConfig default (min_crossings=2) is in force.

    def test_appconfig_default_applies_gate_a_drops_held_shuttle():
        """(a)+(b)+(c)+(e): with the DEFAULT AppConfig (no overrides), Gate A is on ? the
        held-shuttle window (net_crossings<2) is DROPPED from the handoff while the real rally
        (>=2) is KEPT, and the persisted candidate superset is unchanged (both windows persisted)."""
        db, provider, _res = _run(config=AppConfig())
        # (e) persisted superset unchanged: both windows still written to candidate_segments.
        spans = [(kw["start_time"], kw["end_time"]) for _a, kw in db.add_candidate_segment.call_args_list]
        assert spans == [(1.0, 4.0), (10.0, 12.0)]
        # (b)+(c) only the high-crossing rally window A reaches the AI handoff; held window B dropped.
        assert provider.analyze_video.call_count == 1


    def test_appconfig_gate_a_keeps_real_rally():
        """(c) sanity: a config whose ONLY window is the real rally still hands it off under the
        default gate (the gate keeps >=2-crossing windows, it doesn't drop everything)."""
        provider = MagicMock()
        provider.analyze_video.return_value = ([{"start_time": 0.5, "end_time": 2.0}], {})
        db = MagicMock()
        db.add_candidate_segment.side_effect = [101, 102, 103, 104]
```

```
    db.add_play_segment.return_value = 200
    rally_only = MagicMock()
    rally_only.healthcheck.return_value = (True, "env ok")
    rally_only.run_predict.return_value = "out.csv"
    rally_only.parse_trajectory_csv.return_value = _points()
    rally_only.trajectory_to_action_windows.return_value = [dict(WIN_A)]   # rally only
    with patch.object(FusionSegmenter, "_build_runner", return_value=(rally_only, {"weights_path": "w"})), \
         patch.object(FusionSegmenter, "_probe_fps_width", return_value=(30.0, 1280.0)), \
         patch("backend.pipeline.segmenters.trajectory_hybrid.get_video_duration", return_value=30.0), \
         patch("backend.pipeline.segmenters.trajectory_hybrid.extract_video_chunk", return_value=True), \
         patch("backend.pipeline.segmenters.trajectory_hybrid.provider_registry") as preg, \
         patch("os.environ.get", return_value="dummy_key"):
        preg.get.return_value = provider
        FusionSegmenter(db, AppConfig()).process_video("v1", "clip.mp4")
    assert provider.analyze_video.call_count == 1       # the real rally is KEPT by default


def test_gate_a_reversible_via_min_crossings_zero():
    """(d) reversibility: min_crossings=0 restores the prior (ungated) behaviour even on the
    AppConfig-driven served path ? BOTH windows reach the handoff again."""
    cfg = AppConfig.model_validate({"indexing": {"fusion": {"min_crossings": 0}}})
    assert cfg.indexing.fusion.min_crossings == 0
    db, provider, _res = _run(config=cfg)
    assert len(_stats_calls(db)) == 2                   # superset still persisted
    assert provider.analyze_video.call_count == 2       # ungated: both windows handed off


def test_gate_a_default_persists_full_superset():
    """(e) explicit: the default gate filters ONLY the handoff ? every candidate window
    (including the held-shuttle one it drops) is still persisted with its net_crossings stat."""
    db, _provider, _res = _run(config=AppConfig())
    stats = _stats_calls(db)
    assert len(stats) == 2                              # both persisted despite the gate
    assert stats[0]["net_crossings"] >= 2               # rally
    assert stats[1]["net_crossings"] == 0.0             # held shuttle ? persisted, just not handed off


def test_person_cue_adds_features_when_enabled():
    # Two in-court players, each shifting between two sampled frames ? exercises the REAL
    # glue (in-court counts -> players_in_court; track displacement -> mean_player_motion),
    # not just key-presence.
    fake_pd = MagicMock()
    fake_pd.detections_per_frame.return_value = {
        "frames": {
            30: [(1, (10, 10, 30, 40)), (2, (60, 10, 80, 40))],
            33: [(1, (16, 10, 36, 40)), (2, (66, 10, 86, 40))],
        },
        "fps": 30.0, "frame_width": 1000.0, "frame_height": 1000.0,
    }
    with patch("backend.pipeline.detectors.person_detector.PersonDetector", return_value=fake_pd):
        db, _provider, _res = _run(indexing={"fusion": {"cue_flags": {"person": True}}})
```

```
    stats = _stats_calls(db)
    assert all(all(k in s for k in PERSON_FEATURE_NAMES) for s in stats)
    assert all("net_crossings" in s for s in stats)
    assert stats[0]["players_in_court"] == 2.0           # 2 in-court players every sampled frame
    assert stats[0]["mean_player_motion"] > 0.0          # both tracks moved -> non-zero motion
    fake_pd.detections_per_frame.assert_called()         # the person detector WAS used


def test_gate_b_drops_low_player_window_from_handoff():
    """Gate B: with the person cue on and min_players=2, a window whose in-court player
    count is below 2 is dropped from the handoff ? but still persisted (superset)."""
    def _side(video_path, start_f, end_f, frame_skip):
        if start_f >= 300:        # window B [10,12]s -> start frame ~300: only 1 player
            frames = {start_f: [(1, (10, 10, 30, 40))]}
        else:                     # window A [1,4]s -> start frame ~30: 2 players
            frames = {start_f: [(1, (10, 10, 30, 40)), (2, (60, 10, 80, 40))]}
        return {"frames": frames, "fps": 30.0, "frame_width": 1000.0, "frame_height": 1000.0}

    fake_pd = MagicMock()
    fake_pd.detections_per_frame.side_effect = _side
    with patch("backend.pipeline.detectors.person_detector.PersonDetector", return_value=fake_pd):
        db, provider, _res = _run(indexing={"fusion": {"cue_flags": {"person": True}, "min_players": 2}})
    stats = _stats_calls(db)
    assert len(stats) == 2                               # superset persisted (both)
    assert provider.analyze_video.call_count == 1        # only window A (>=2 players) handed off
    assert stats[0]["players_in_court"] >= 2 and stats[1]["players_in_court"] < 2


def test_build_runner_selects_substrate():
    """_build_runner picks the WASB or TrackNet runner by indexing.fusion.substrate."""
    seg = FusionSegmenter(MagicMock(), {"indexing": {}})
    with patch("backend.pipeline.segmenters.fusion_hybrid.WasbRunner") as W, \
         patch("backend.pipeline.segmenters.fusion_hybrid.WasbConfig") as WC, \
         patch("backend.pipeline.segmenters.fusion_hybrid.TrackNetRunner") as T, \
         patch("backend.pipeline.segmenters.fusion_hybrid.TrackNetConfig") as TC:
        WC.from_indexing_cfg.return_value = MagicMock(weights_path="w")
        TC.from_indexing_cfg.return_value = MagicMock(weights_path="t", queue_length=5)
        _runner_w, extras_w = seg._build_runner({"fusion": {"substrate": "wasb"}})
        _runner_t, extras_t = seg._build_runner({"fusion": {"substrate": "tracknet"}})
        _runner_d, extras_d = seg._build_runner({})      # default -> wasb
    assert W.called and extras_w["fusion_substrate"] == "wasb"
    assert T.called and extras_t["fusion_substrate"] == "tracknet"
    assert extras_t["queue_length"] == 5
    assert extras_d["fusion_substrate"] == "wasb"

```

38. assistant (2026-06-14T10:51:43.547Z)

Now let me see the pre-#146 version of this file to restore the changed parts faithfully.

39. user (2026-06-14T10:51:45.071Z)

```
"""FusionSegmenter tests (mocked runner/provider/db ? no GPU/WSL).

Proves: net_crossings is persisted; the person cue is OFF by default (no person-feature
keys, PersonDetector never built); the persisted candidate set is the full superset of the
substrate windows; served Gate A drops sub-threshold windows from the AI handoff while
persistence keeps the superset; and the person cue, when enabled, adds the person features.
"""
```

```
from types import SimpleNamespace
from unittest.mock import MagicMock, patch

from backend.pipeline.detectors.fusion_features import PERSON_FEATURE_NAMES
from backend.pipeline.segmenters.fusion_hybrid import FusionSegmenter
```

Window A [1,4]s: shuttle alternates across the net every frame -> many crossings (a ra

Window B [10,12]s: shuttle sits at the net level (held) -> 0 crossings. fps=30.

```
WIN_A = {"start": 1.0, "end": 4.0, "active_frame_count": 90, "peak_velocity": 0.4,
         "mean_velocity": 0.2, "track_id_count": 2, "chunk_index": 0}
WIN_B = {"start": 10.0, "end": 12.0, "active_frame_count": 60, "peak_velocity": 0.1,
         "mean_velocity": 0.05, "track_id_count": 1, "chunk_index": 0}


def _points():
    pts = []
    for f in range(30, 120):       # window A frames: y flips 100<->900 around the net (~500)
        pts.append(SimpleNamespace(frame=f, visible=True, x=500.0, y=100.0 if f % 2 == 0 else
900.0))
    for f in range(300, 360):      # window B frames: y == net level -> deadband -> no crossings
        pts.append(SimpleNamespace(frame=f, visible=True, x=500.0, y=500.0))
    return pts


def _runner():
    r = MagicMock()
    r.healthcheck.return_value = (True, "env ok")
    r.run_predict.return_value = "out.csv"
    r.parse_trajectory_csv.return_value = _points()
    r.trajectory_to_action_windows.return_value = [dict(WIN_A), dict(WIN_B)]
    return r


def _run(indexing=None, provider_segments=None):
    provider = MagicMock()
    provider.analyze_video.return_value = (provider_segments or [{"start_time": 0.5, "end_time":
2.0}], {})
    db = MagicMock()
    db.add_candidate_segment.side_effect = [101, 102, 103, 104]
    db.add_play_segment.return_value = 200

    with patch.object(FusionSegmenter, "_build_runner", return_value=(_runner(),
{"weights_path": "w"})), \
         patch.object(FusionSegmenter, "_probe_fps_width", return_value=(30.0, 1280.0)), \
         patch("backend.pipeline.segmenters.trajectory_hybrid.get_video_duration",
return_value=30.0), \
         patch("backend.pipeline.segmenters.trajectory_hybrid.extract_video_chunk",
return_value=True), \
         patch("backend.pipeline.segmenters.trajectory_hybrid.provider_registry") as preg, \
         patch("os.environ.get", return_value="dummy_key"):
        preg.get.return_value = provider
        seg = FusionSegmenter(db, {"indexing": indexing or {}})
        res = seg.process_video("v1", "clip.mp4")
        return db, provider, res


def _stats_calls(db):
    """Persisted stats dicts, in call order."""
    return [kw["stats"] for _a, kw in db.add_candidate_segment.call_args_list]


def test_net_crossings_persisted_and_person_off_by_default():
    db, _provider, _res = _run()
```

```
    stats = _stats_calls(db)
    assert len(stats) == 2                              # both windows persisted (superset)
    assert all("net_crossings" in s for s in stats)
    # Window A (rally) crosses many times; window B (held) zero.
    assert stats[0]["net_crossings"] >= 2
    assert stats[1]["net_crossings"] == 0.0
    # Person cue OFF by default -> NONE of the person features present.
    assert all(not any(k in s for k in PERSON_FEATURE_NAMES) for s in stats)
    # The 4 base motion keys are still there.
    assert all({"peak_velocity", "mean_velocity", "active_frame_count", "track_id_count"} <=
set(s)
               for s in stats)


def test_persisted_candidates_are_superset_of_substrate_windows():
    db, _provider, _res = _run()
    spans = [(kw["start_time"], kw["end_time"]) for _a, kw in
db.add_candidate_segment.call_args_list]
    assert spans == [(1.0, 4.0), (10.0, 12.0)]          # exactly the substrate windows,
unfiltered


def test_default_hands_off_all_windows():
    _db, provider, _res = _run()                        # min_crossings default 0 -> no gating
    assert provider.analyze_video.call_count == 2


def test_gate_a_drops_low_crossing_window_from_handoff_only():
    db, provider, _res = _run(indexing={"fusion": {"min_crossings": 2}})
    # Persistence is still the full superset (offline substrate intact)...
    assert len(_stats_calls(db)) == 2
    # ...but only the high-crossing window A reaches the AI handoff.
    assert provider.analyze_video.call_count == 1


def test_person_cue_adds_features_when_enabled():
    # Two in-court players, each shifting between two sampled frames ? exercises the REAL
    # glue (in-court counts -> players_in_court; track displacement -> mean_player_motion),
    # not just key-presence.
    fake_pd = MagicMock()
    fake_pd.detections_per_frame.return_value = {
        "frames": {
            30: [(1, (10, 10, 30, 40)), (2, (60, 10, 80, 40))],
            33: [(1, (16, 10, 36, 40)), (2, (66, 10, 86, 40))],
        },
        "fps": 30.0, "frame_width": 1000.0, "frame_height": 1000.0,
    }
    with patch("backend.pipeline.detectors.person_detector.PersonDetector",
return_value=fake_pd):
        db, _provider, _res = _run(indexing={"fusion": {"cue_flags": {"person": True}}})
    stats = _stats_calls(db)
    assert all(all(k in s for k in PERSON_FEATURE_NAMES) for s in stats)
    assert all("net_crossings" in s for s in stats)
    assert stats[0]["players_in_court"] == 2.0          # 2 in-court players every sampled
frame
    assert stats[0]["mean_player_motion"] > 0.0         # both tracks moved -> non-zero motion
    fake_pd.detections_per_frame.assert_called()        # the person detector WAS used


def test_gate_b_drops_low_player_window_from_handoff():
```

40. assistant (2026-06-14T10:51:48.851Z)

I have the pre-#146 version. The cleanest, most faithful approach is to restore `tests/test_fusion_hybrid.py` to its pre-#146 form via `git checkout`, since #146 only added

Gate-A-default tests and modified the docstring/`_run` helper. This avoids leaving stale AppConfig-default tests that assert the reverted behavior. Let me restore it from `570cf2b^`.

41. user (2026-06-14T10:51:50.692Z)

restored

42. assistant (2026-06-14T10:51:55.396Z)

`tests/test_fusion_hybrid.py` is now byte-identical to its pre-#146 state.

Now I have a decision on test_config.py: my rewrite preserves additional coverage (opt-in + served-default=gemini tests). That's reasonable and not stale. But let me reconsider ? the brief says fix the tests "back to the restored defaults". The pre-#146 test_config.py had no served-default test at all. My added `test_served_default_segmenter_is_gemini` is a useful guard against recurrence and asserts the restored value, which aligns with the owner's "ensure we do not regress" intent. I'll keep it but make sure it's clean. Let me view the current test_config.py section to confirm.

43. user (2026-06-14T10:51:56.016Z)

```

def test_fusion_config_defaults_reproduce_control():
    """The fusion block defaults must be no-ops: wasb substrate, no cues, no gating, no mask.

    Gate A (min_crossings) defaults to 0 = OFF: the #146 served-default flip was REVERTED
    (eval?serve ? the +0.074 was on the harness's recall-oriented PROPOSAL windowing; on the
    served COARSE windowing Gate A is neutral/negative, ?0.008, dropping real rallies). See
    docs/QUALITY_ITERATIONS.md and the served litmus in tests/test_served_gate_a_regression.py.
    """
    cfg = load_config("/nonexistent/path/12345/config.json")
    f = cfg.indexing.fusion
    assert f.substrate == "wasb"
    assert f.cue_flags == {}            # person cue OFF
    assert f.min_crossings == 0         # Gate A OFF (default) ? #146 served flip reverted
    assert f.min_players == 0           # Gate B OFF
    assert f.court_polygon is None      # no mask
    assert f.velocity_threshold == 0.02


def test_fusion_config_gate_a_opt_in_via_min_crossings():
    """Gate A is opt-in: setting min_crossings explicitly enables it (the eval-only operating
    point), while the served default stays OFF (#146 flip reverted)."""
    with tempfile.TemporaryDirectory() as td:
        p = os.path.join(td, "optin.json")
        with open(p, "w") as fh:
            json.dump({"indexing": {"fusion": {"min_crossings": 2}}}, fh)
        cfg = load_config(p)
        assert cfg.indexing.fusion.min_crossings == 2   # Gate A ON only when explicitly set


def test_served_default_segmenter_is_gemini():
    """The shipped config.json served default is the prior detector (gemini) ? the #146 flip
    to fusion_hybrid was REVERTED as a confirmed served regression (eval?serve)."""
    repo_root = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
```

```
        cfg = load_config(os.path.join(repo_root, "config.json"))
        assert cfg.indexing.default_segmenter == "gemini"
        # ...and the served fusion default keeps Gate A OFF (no served gating).
        assert cfg.indexing.fusion.min_crossings == 0


    def test_fusion_config_file_override():
        with tempfile.TemporaryDirectory() as td:
```

44. assistant (2026-06-14T10:51:58.939Z)

test_config.py looks good. Now the docs. Let me read the relevant section of `docs/QUALITY_ITERATIONS.md` to amend the Gate-A entry to REVERTED. First let me check if a concurrent session already added a served-finding entry.

45. user (2026-06-14T10:51:59.785Z)

```
| + person (P3) | 0.286 | **?0.021** (?6.7%) | [?0.165, +0.162] | 2W/4L (p=0.688) | **No** |
| + person + audio (P4 incr.) | 0.366 | **+0.079** (+27.7%) | [?0.054, +0.203] | 4W/2L (p=0.688) | **No** |

**Read honestly:** neither cue clears the promotion bar (both CIs span 0, both sign tests n.s.) at n=6 ? **both stay
default-OFF** (promote-only-on-lift). **Person** adds noise (no court polygons ? `players_in_court` counts spectators);
per-video it's a wash (testlarge +0.40, gbaaddy ?0.22). **Audio** is the more promising lead ? +0.079 / +27.7%, wins 4/6
(largetest +0.27, mahadevpura +0.27), and shuttle+person+audio (0.366) lands **above** shuttle-only (0.307) ? but n=6 can't
confirm it. **Next:** more matches (raise N until the sign test can resolve ~+0.08), and measure audio on shuttle-only
*directly* (without the person handicap). Recorded in `eval_baselines/experiment_leaderboard.json`. **Negative/uncertain
results are kept on purpose ? this is the honest ablation backbone for the paper aim.**

### Gate-A flipped to the SERVED DEFAULT (owner-approved) *(2026-06-14)*
**Gate-A (net-crossings ? 2) is now the DEFAULT detection behaviour.** The owner approved promoting the
validated held-shuttle guard from "recommended config" to the live served default. The flip is **two minimal,
config-reversible changes**:
- `backend/config/models.py` ? `FusionConfig.min_crossings` default **0 ? 2** (Gate A on wherever
  `fusion_hybrid` serves).
- `config.json` ? `indexing.default_segmenter` **`gemini` ? `fusion_hybrid`** (+ an explicit `indexing.fusion`
  block making the served operating point visible). With default config, `FusionSegmenter` reproduces the WASB
  substrate's seeding/handoff exactly *except* that sub-2-crossing windows are dropped from the AI handoff ? the
  held-shuttle / one-sided-feed / warm-up false-positive fix.

**Reversible ? rollback is a single flag** (no code change): set `indexing.fusion.min_crossings: 0` to disable
Gate A (ungated handoff, reproduces the substrate byte-for-byte), or `indexing.default_segmenter: wasb_hybrid`
to revert the served segmenter entirely. **Gate B stays OFF** (`min_players = 0`, person cue OFF). **The
persisted candidate superset is unchanged** ? gating filters only the AI handoff, so the offline harness can
still re-score any variant; the offline ablation harness is untouched (it sets `Variant.min_crossings`
explicitly and never reads `FusionConfig`). The validation evidence is unchanged from the promotion below
```

(?F1 +0.074, 6/6, p=0.031, CI [+0.042, +0.105], over-seg 1.68?1.01) plus the GX010125 45?27 corroboration.
Recorded: a `promotion` record appended to `eval_baselines/experiment_leaderboard.json`. ? **Honesty
boundary:** the offline ?F1 is on the recall-oriented *proposal* candidate set, and the production fusion
segmenter's served path is still only mock-tested on real video (P2). The gate's effect is **byte-reversible**,
so this ships behind a one-flag rollback; a golden *served* regression on the fusion path remains a worthwhile
optional follow-up (owner-gated GPU+Gemini run) to confirm no served-side regression, but it does not block the
flip ? the offline 6/6 + p=0.031 ablation and the unit tests are the gate.

Acting on the first A/B: Gate-A PROMOTED + learned variant FIXED + GX010125 corroboration *(2026-06-14)*
Gate-A (net-crossings ? 2) ? PROMOTED (first cue promoted via the harness). The measured win
(?F1 +0.074, **6/6 golden matches, sign-test p=0.031**, CI [+0.042,+0.104]; over-seg 1.68?1.01 ? the
[first-A/B archive](archives/quality-iterations/2026-06-first-ab-experiment/README.md)) clears the honest
bar. **Recommended config:** `indexing.fusion.min_crossings = 2` on the fusion path (the eval-only Gate A
folded into production, MULTI_SIGNAL_FUSION_PLAN §3.1). ? The offline win is on the recall-oriented *proposal*
candidate set; flipping the **live served default** stayed owner-gated at the time of this entry ? **now resolved:
the owner approved the flip; see the SERVED-DEFAULT entry above.** **Corroborated on real footage:** offline
re-score of GX010125_1080p's *cached* trajectory (no GPU, isolated `output/GX010125_run/`) ? 45 candidates ?
CONTROL 45 rallies vs **Gate-A 27**; Gate-A dropped **18 low-crossing windows (~1.7 min)** of likely
held-shuttle/feed FPs (no golden labels for this footage ? divergence, not F1).

Learned variant FIXED ? [#130](https://github.com/avidullu/badminton-highlight-indexer/pull/130). The
first A/B's learned logistic lost to CONTROL and zeroed `testlarge` ? a fixed-0.5-cutoff artifact. Added
additive/default-OFF `Variant.tune_threshold` (operating point chosen on the **train** fold) + `calibrate`
(Platt/isotonic, C2). Validated on the n=6 golden set: **mean F1 0.307 ? 0.423 (+0.116)**, testlarge
0.000 ? 0.300, and the learned variant now **beats CONTROL (0.388)**. Threshold-in-LOVO is the big lever;
calibration is available too.

Next: a real fusion-path run to confirm Gate-A served-side; per-video output layout
(PER_VIDEO_OUTPUT_LAYOUT.md) so each video's artifacts are self-contained.

First end-to-end A/B on the golden corpus (n=6) ? **ARCHIVED** *(2026-06-14)*
First real use of the harness + C1?C4 helpers, 100% offline (re-scored persisted WASB trajectories, no GPU).

46. assistant (2026-06-14T10:52:05.997Z)

I can see the `master` state. The brief mentions a concurrent session may have already added a served-finding entry. Let me check the current origin/master and the live worktree's QUALITY_ITERATIONS.md to see if anything was added beyond what I'm seeing. The file I'm reading is from the worktree (origin/master), so this is current. There's no separate "served-finding entry" yet in this doc ? only the #146 promotion entry. PR #148 (which contains the served finding) only adds two files, not docs. So no doc duplication concern here on master.

Now let me amend the Gate-A SERVED-DEFAULT entry to read REVERTED. I'll replace the entry with an honest REVERTED entry.

47. user (2026-06-14T10:52:31.096Z)

The file C:/Users/avidu/Projects/bhi-revert/docs/QUALITY_ITERATIONS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

48. assistant (2026-06-14T10:52:34.667Z)

Now the entry below (the PROMOTED entry) still says "now resolved: the owner approved the flip; see the SERVED-DEFAULT entry above." That's now stale/contradictory since the flip was reverted. Let me amend that pointer to reflect the revert.

49. user (2026-06-14T10:52:39.567Z)

The file C:/Users/avidu/Projects/bhi-revert/docs/QUALITY_ITERATIONS.md has been updated successfully. (file state is current in your context ? no need to Read it back)

50. assistant (2026-06-14T10:52:42.397Z)

Now append a "reverted (eval?serve)" note to the leaderboard's #146 promotion record. Let me view the current state of that record in the worktree.

51. user (2026-06-14T10:52:43.275Z)


```
            "p_value": 0.6875
          }
        }
      },
      {
        "timestamp": "2026-06-14T00:00:00+00:00",
        "iou": 0.5,
        "label_set_hash": "50d98ffbc370",
        "num_videos": 6,
        "promotion": {
          "cue": "gate_a_net_crossings",
          "promoted_to": "served_default",
          "decision": "PROMOTED ? Gate A (net_crossings>=2) flipped to the live served default",
          "served_change": "config.json indexing.default_segmenter gemini->fusion_hybrid; FusionConfig.min_crossings default 0->2",
          "reversible": "indexing.fusion.min_crossings=0 (Gate-A rollback) or indexing.default_segmenter=wasb_hybrid (full revert)",
          "gate_b": "stays OFF (min_players=0, person cue OFF)",
          "owner_approved": true,
          "real_video_corroboration": "GX010125_1080p offline re-score: 45 candidates -> CONTROL 45 rallies vs Gate-A 27 (dropped 18 low-crossing held-shuttle/feed windows, ~1.7min)",
          "over_seg_segment_count_ratio": {"before": 1.68, "after": 1.01},
          "source_archive": "docs/archives/quality-iterations/2026-06-first-ab-experiment/README.md"
        },
        "variant_a": {
          "name": "control_shuttle_proposals",
          "spec_hash": "control",
          "aggregate": {}
        },
        "variant_b": {
          "name": "gate_a_min_crossings_2",
          "spec_hash": "gateA_mc2",
          "aggregate": {}
        },
        "delta": {
          "f1": 0.074
```

```
      },
      "honest_delta_f1": {
        "mean_delta": 0.074,
        "ci_low": 0.042,
        "ci_high": 0.105,
        "n": 6.0,
        "ci_excludes_zero": true,
        "sign_test": {
          "wins": 6.0,
          "losses": 0.0,
          "ties": 0.0,
          "n_effective": 6.0,
          "p_value": 0.031
        }
      }
    }
  ]
```


52. assistant (2026-06-14T10:52:46.252Z)

I'll add a `reverted` field to the promotion record. Note the file ends with no trailing newline (the original had `\ No newline at end of file`). Let me add the note inside the `promotion` object.

53. user (2026-06-14T10:52:53.554Z)

The file C:/Users/avidu/Projects/bhi-revert/eval_baselines/experiment_leaderboard.json has been updated successfully. (file state is current in your context ? no need to Read it back)

54. assistant (2026-06-14T10:52:55.643Z)

Let me validate the JSON is still well-formed.

55. user (2026-06-14T10:52:56.994Z)

JSON OK
config.json OK

56. assistant (2026-06-14T10:52:59.651Z)

Both JSON files are valid. Now let me run the linters and the full test suite. First, let me check how tests are run in this repo and run ruff/mypy/pytest.

57. user (2026-06-14T10:53:02.577Z)

All checks passed!

58. user (2026-06-14T10:53:33.580Z)

```
backend\annotations\base.py:85: note: By default the bodies of untyped functions are not checked, consider using --check-untyped-defs  [annotation-unchecked]
backend\sports\__init__.py:7: note: By default the bodies of untyped functions are not checked, consider using --check-untyped-defs  [annotation-unchecked]
backend\pipeline\validators\base.py:36: note: By default the bodies of untyped functions are not checked, consider using --check-untyped-defs  [annotation-unchecked]
backend\providers\__init__.py:7: note: By default the bodies of untyped functions are not checked, consider using --check-untyped-defs  [annotation-unchecked]
Success: no issues found in 91 source files
```

59. assistant (2026-06-14T10:53:35.865Z)

Ruff and mypy are clean. Let me check the CI config to see exactly how mypy is invoked (the brief says mypy green), then run the full test suite.

60. user (2026-06-14T10:54:27.913Z)

```
========================================================================
 Full test suite:  C:\Users\avidu\AppData\Local\Programs\Python\Python313\python.exe -m pytest
-q
 obsreport (optional reporting dep): available
========================================================================
....................ss.................................................. [ 11%]
........................................................................ [ 23%]
........................................................................ [ 35%]
........................................................................ [ 47%]
........................................................................ [ 59%]
........................................................................ [ 71%]
........................................................................ [ 83%]
........................................................................ [ 94%]
...............................                                          [100%]
605 passed, 2 skipped in 49.01s
========================================================================
 RESULT: PASS - all tests green
========================================================================
```

61. assistant (2026-06-14T10:54:32.274Z)

Full suite: 605 passed, 2 skipped (the 2 skips are golden-data-gated, expected offline). Note this worktree does not have PR #148's tests yet, so the count is lower than #148's 609 ? that's expected.

Now let me review the full diff against origin/master before committing, to confirm only the intended changes are present.

62. user (2026-06-14T10:54:33.368Z)

```
 backend/config/models.py                      | 33 +++---------
 config.json                                   |  8 +--
 docs/QUALITY_ITERATIONS.md                    | 63 +++++++++++++---------
 eval_baselines/experiment_leaderboard.json |  1 +
 tests/test_config.py                          | 38 +++++++------
 tests/test_fusion_hybrid.py                   | 86 ++----------------------------
 6 files changed, 72 insertions(+), 157 deletions(-)
---FILES---
 M backend/config/models.py
 M config.json
 M docs/QUALITY_ITERATIONS.md
 M eval_baselines/experiment_leaderboard.json
 M tests/test_config.py
M  tests/test_fusion_hybrid.py
```

63. assistant (2026-06-14T10:54:36.495Z)

Exactly the 6 intended files. Let me do a final review of the docs diff to make sure it reads cleanly, then commit.

64. user (2026-06-14T10:54:37.242Z)

```
diff --git a/backend/config/models.py b/backend/config/models.py
index b18693e..ff94380 100644
--- a/backend/config/models.py
+++ b/backend/config/models.py
@@ -78,26 +78,12 @@ class WasbIndexConfig(BaseModel):
 class FusionConfig(BaseModel):
     """Config for the multi-signal `fusion_hybrid` segmenter (MULTI_SIGNAL_FUSION_PLAN P2).

-    **Gate A (the held-shuttle guard) is ON by default** (``min_crossings=2``): a window
-    must show the shuttle crossing the net >= 2 times to reach the AI handoff, dropping the
```

```
-    held-shuttle / one-sided-feed / warm-up false positives. This is the promoted, validated
-    operating point (?F1 +0.074, 6/6 golden, sign-test p=0.031, CI [+0.042,+0.105]; over-seg
-    1.68?1.01 ? see docs/QUALITY_ITERATIONS.md). It only takes effect when the served segmenter
-    is ``fusion_hybrid`` (config.json ``indexing.default_segmenter``); the offline experiment
-    harness is unaffected (it sets ``Variant.min_crossings`` explicitly, never reading this).
-
-    **Reversible ? rollback is a single flag:** set ``indexing.fusion.min_crossings: 0`` to
-    restore the pre-Gate-A (ungated) handoff, or ``indexing.default_segmenter: wasb_hybrid``
-    to revert the served segmenter entirely. With ``min_crossings=0`` the fusion segmenter
-    reproduces the substrate (wasb_hybrid) byte-for-byte.
-
-    The person cue (Gate B) stays OFF by default: the fusion segmenter persists the shuttle
-    net-crossing count always, and ? only when ``cue_flags['person']`` is true ? the person-cue
-    features; ``min_players`` (Gate B) defaults to 0 = OFF. The court mask is optional:
-    ``court_polygon=None`` ? Gate B counts every detection (player-count-only); supplied ?
-    off-court persons (spectators / adjacent courts) are excluded. **The persisted candidate
-    set is the full superset regardless of any gate** ? gating only filters the AI handoff, so
-    the offline harness can re-score any variant.
+    Every default reproduces control behaviour: the fusion segmenter persists the
+    shuttle net-crossing count (Gate A signal) and ? only when ``cue_flags['person']`` is
+    true ? the person-cue features, but it does NOT gate the served handoff unless a
+    threshold is raised (``min_crossings``/``min_players`` default 0 = OFF). The court mask
+    is optional: ``court_polygon=None`` ? Gate B counts every detection (player-count-only);
+    supplied ? off-court persons (spectators / adjacent courts) are excluded.
     """
     # Which trajectory substrate seeds the windows (shuttle stays primary).
     substrate: Literal["wasb", "tracknet"] = "wasb"
@@ -109,11 +95,8 @@ class FusionConfig(BaseModel):
     # enabled ? so the default path never imports torch / touches the GPU.
     cue_flags: dict[str, bool] = Field(default_factory=dict)
     # Served Gate A: drop windows whose shuttle net-crossings < this from the AI handoff.
-    # Default 2 = the PROMOTED held-shuttle guard (a real rally crosses the net >=2x; a single
-    # service feed / held shuttle crosses ~once). Set 0 to turn the gate OFF (ungated handoff;
-    # reproduces the substrate exactly). Persisted candidates are ALWAYS the full superset for
-    # offline re-scoring, regardless of this threshold.
-    min_crossings: int = Field(default=2, ge=0)
+    # 0 = OFF (no gating; persisted candidates are always the full superset for offline re-
scoring).
+    min_crossings: int = Field(default=0, ge=0)
     # Served Gate B: when the person cue is on, drop windows with median in-court players
     # < this. 0 = OFF.
     min_players: int = Field(default=0, ge=0)
diff --git a/config.json b/config.json
index 3107f70..b72a93a 100644
--- a/config.json
+++ b/config.json
@@ -23,7 +23,7 @@
         }
     },
     "indexing": {
-        "default_segmenter": "fusion_hybrid",
+        "default_segmenter": "gemini",
         "use_gpu": true,
         "motion_threshold": 0.08,
         "min_segment_duration": 3.0,
@@ -63,12 +63,6 @@
             "wsl_stage_dir": "~/clips",
             "timeout_sec": 1800,
             "velocity_threshold": 0.02
-        },
-        "fusion": {
-            "substrate": "wasb",
-            "min_crossings": 2,
-            "min_players": 0,
-            "cue_flags": {}
```

```
        }
    },
    "output": {
```

65. assistant (2026-06-14T10:54:40.711Z)

```
The code/config diff is exactly the inverse of #146 (restoring pre-#146 state). Now let me
verify branch safety before committing, then commit with explicit paths.
```

66. user (2026-06-14T10:54:44.251Z)

```
---branch---
fix/revert-gate-a-default
---HEAD---
e75ec35 feat(eval): LOSO/LOGO ablation runner ? per-signal marginal ?F1 (Phase 0) (#147)
---commits ahead of origin/master (should be empty before commit)---
```

67. assistant (2026-06-14T10:54:46.925Z)

```
On the correct branch, HEAD = origin/master tip (e75ec35), no sibling commits crept in. Now let
me stage explicit paths and commit.
```

68. user (2026-06-14T10:54:48.672Z)

```
M  backend/config/models.py
M  config.json
M  docs/QUALITY_ITERATIONS.md
M  eval_baselines/experiment_leaderboard.json
M  tests/test_config.py
M  tests/test_fusion_hybrid.py
```

69. user (2026-06-14T10:55:01.942Z)

```
[fix/revert-gate-a-default 7883a7c] revert(fusion): undo the Gate-A served-default flip (#146) ?
confirmed served regression (eval?serve)
 6 files changed, 72 insertions(+), 157 deletions(-)
```

70. user (2026-06-14T10:55:06.377Z)

```
---only my commit?---
7883a7c revert(fusion): undo the Gate-A served-default flip (#146) ? confirmed served regression
(eval?serve)
---HEAD is mine---
7883a7c revert(fusion): undo the Gate-A served-default flip (#146) ? confirmed served regression
(eval?serve)
```

71. assistant (2026-06-14T10:55:10.958Z)

```
`git log origin/master..HEAD` lists only my single commit. Now let me push and open the PR.
```

72. user (2026-06-14T10:55:13.920Z)

```
remote:
remote: Create a pull request for 'fix/revert-gate-a-default' on GitHub by visiting:
remote:      https://github.com/avidullu/badminton-highlight-indexer/pull/new/fix/revert-gate-a-
default
remote:
branch 'fix/revert-gate-a-default' set up to track 'origin/fix/revert-gate-a-default'.
To https://github.com/avidullu/badminton-highlight-indexer.git
 * [new branch]      fix/revert-gate-a-default -> fix/revert-gate-a-default
```

73. user (2026-06-14T10:55:34.734Z)

https://github.com/avidullu/badminton-highlight-indexer/pull/151

74. assistant (2026-06-14T10:55:37.668Z)

Revert PR is #151. Now let me wait for CI to pass on #151, and re-check #148's status. Let me poll the revert PR's checks.

75. user (2026-06-14T10:55:40.858Z)

Exit code 8
{"mergeStateStatus":"UNSTABLE","mergeable":"MERGEABLE","number":151}
---checks---
Full test suite (offline, mocked)   pending 0   https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27496605671/job/81271722356
Mypy (lenient baseline) pending 0   https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27496605671/job/81271722415
Ruff lint   pending 0   https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27496605671/job/81271722358
Syntax sweep + import smoke (no GPU stack)   pending 0
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27496605671/job/81271722367

76. assistant (2026-06-14T10:55:43.765Z)

CI is running. Let me load the Monitor tool to wait for CI to complete rather than polling in a sleep loop.

77. assistant (2026-06-14T10:55:47.711Z)

I'll use Bash with `run_in_background` to wait for CI to reach a terminal state on #151 (single notification when it completes).

78. user (2026-06-14T10:55:50.527Z)

Command running in background with ID: b8m8fv2vy. Output is being written to: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\985cc286-9e1e-4217-87c3-2d418bc4fa9c\tasks\b8m8fv2vy.output. You will be notified when it completes. To check interim output, use Read on that file path.

79. assistant (2026-06-14T10:55:54.279Z)

While CI runs on #151, let me double check that PR #148 is still green/mergeable and confirm there are no doc/file overlaps between #148 and my revert that would cause a conflict when both land.

80. user (2026-06-14T10:55:56.226Z)

{"files":[{"path":"backend/eval/served_gate_a.py","additions":199,"deletions":0,"changeType":"ADDED"},{"path":"tests/test_served_gate_a_regression.py","additions":160,"deletions":0,"changeType":"ADDED"}],"headRefName":"feat/served-gate-a-regression","mergeStateStatus":"UNKNOWN","mergeable":"UNKNOWN"}

81. assistant (2026-06-14T10:56:00.696Z)

#148 only adds two new files (`backend/eval/served_gate_a.py`, `tests/test_served_gate_a_regression.py`) ? zero overlap with my revert's 6 files, so no conflict. The mergeStateStatus is "UNKNOWN" right now (GitHub recomputing because master/PR state is in flux). Let me re-check #148 in a moment. First let me check the background job's interim output.

82. user (2026-06-14T10:56:01.395Z)

<system-reminder>Warning: the file exists but is shorter than the provided offset (1). The file has 1 lines.</system-reminder>

83. assistant (2026-06-14T10:56:04.194Z)

Still waiting on CI. Let me re-check #148's mergeability now that GitHub may have recomputed.

84. user (2026-06-14T10:56:07.551Z)

Full test suite (offline, mocked)	pass	2m2s	https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27496356623/job/81271029709
Mypy (lenient baseline)	pass	32s	https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27496356623/job/81271029652
Ruff lint	pass	9s	https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27496356623/job/81271029679
Syntax sweep + import smoke (no GPU stack)	pass	17s	https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27496356623/job/81271029666
---merge state---
{"mergeStateStatus":"CLEAN","mergeable":"MERGEABLE"}

85. assistant (2026-06-14T10:56:14.197Z)

#148 is green (all 4 checks pass), CLEAN, MERGEABLE. It's additive infra (the served litmus that caught this), consistent with the revert, and doesn't conflict. The brief authorizes merging it on green.

Let me think about merge order. Both PRs target master. #148 is independent (2 new files). My revert #151 is based on the same origin/master. If I merge #148 first, master will move, and #151 (a squash merge) will need to be mergeable on top ? but since the file sets are disjoint, #151 will still merge cleanly. Equally, merging #151 first then #148 works.

Let me merge #148 now (it's green and ready), which also gives my revert a clean target. Per the owner's working-agreement, Claude is authorized to merge its own good-engineering-fix PRs on verified-green CI. #148 is a verification-infra PR (not product/strategy/licensing), and the brief explicitly directs merging it on green.

86. user (2026-06-14T10:56:18.296Z)

(Bash completed with no output)

87. user (2026-06-14T10:56:23.864Z)

{"mergeCommit":{"oid":"bd2a216905abded66ba0202e54a93e3bb84b13d5"},"mergedAt":"2026-06-14T10:56:17Z","state":"MERGED"}
---master log---
bd2a2169 feat(eval): served Gate-A verification + repeatable regression litmus (#146 follow-up) (#148)

88. assistant (2026-06-14T10:56:29.829Z)

#148 is merged (master now at bd2a2169). Now my revert #151 needs to be mergeable on the new master. Since the file sets are disjoint, it should still be CLEAN. Let me check the background CI job and #151's merge state.

89. user (2026-06-14T10:56:31.921Z)

<system-reminder>Warning: the file exists but is shorter than the provided offset (1). The file has 1 lines.</system-reminder>

90. assistant (2026-06-14T10:56:37.143Z)

CI on #151 hasn't reached terminal yet. Let me check its current state directly.

91. user (2026-06-14T10:56:40.137Z)

Mypy (lenient baseline)	pass	26s	https://github.com/avidullu/badminton-highlight-

indexer/actions/runs/27496605671/job/81271722415
Ruff lint	pass	8s	https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27496605671/job/81271722358
Syntax sweep + import smoke (no GPU stack)	pass	15s	https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27496605671/job/81271722367
Full test suite (offline, mocked)	pending	0	https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27496605671/job/81271722356
---merge state---
{"mergeStateStatus":"UNKNOWN","mergeable":"UNKNOWN"}

92. assistant (2026-06-14T10:56:45.292Z)

3 of 4 checks pass; the full test suite is still running. The merge state is UNKNOWN because GitHub is recomputing against the new master (after #148 landed). The background monitor will notify me when CI is terminal. Let me wait for it. In the meantime, let me verify my revert branch still merges cleanly against the updated master by a local check (the file sets are disjoint, but let me confirm there's no conflict).

93. user (2026-06-14T10:56:49.042Z)

```
From https://github.com/avidullu/badminton-highlight-indexer
   8439c24..bd2a216  master     -> origin/master
---can my branch merge cleanly into new master? (merge-tree check)---
NO CONFLICTS
```

94. assistant (2026-06-14T10:56:52.909Z)

No conflicts against the updated master. The revert will merge cleanly. Now let me wait for the background CI monitor to report the full test suite result.

95. user (2026-06-14T10:56:55.351Z)

<system-reminder>Warning: the file exists but is shorter than the provided offset (1). The file has 1 lines.</system-reminder>

96. assistant (2026-06-14T10:57:03.475Z)

Still running. The background job will notify me when CI reaches a terminal state. Let me wait for that notification.

97. user (2026-06-14T10:57:04.064Z)

<system-reminder>Warning: the file exists but is shorter than the provided offset (1). The file has 1 lines.</system-reminder>

98. assistant (2026-06-14T10:57:06.520Z)

The background monitor is still polling. I'll wait for its completion notification rather than polling repeatedly.